Quantum Computing
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Lecture Date: August 25, 2026

Lecture Topic: Efficient Algorithms, Random Sampling, Lower Bound for Deterministic Search, Probability Basics

In this first lecture, we see why randomness and quantum computation matter for two basic problems, primality/factoring and search. We introduce random sampling as the basic primitive of randomized algorithms, prove a lower bound of $n-1$ queries for deterministic search, and review the basics of probability that we need for the rest of the course.

1 Efficient Algorithms and Finding Prime Numbers

Classical computing was deterministic until relatively recently, when randomness was introduced into algorithms. The next step, and the subject of this course, is quantum computing.

Consider the problem of deciding whether a given number n is prime. The input is the binary representation of n, so the *input size* is the number of bits

$$N = \log_2 n,$$

and not n itself.

Definition 1 ▶ Efficient Algorithm

An algorithm is **efficient** if it runs in time polynomial in the input size, i.e., in time $\mathcal{O}\left(N^k\right)$ for some constant k. For primality testing this means time $\mathcal{O}\left((\log_2 n)^k\right)$.

The trivial algorithm, trial division by all numbers up to $\sqrt{n}$, takes $\Theta(\sqrt{n}) = \Theta\left(2^{N/2}\right)$ time, which is exponential in the input size. For a long time no efficient algorithm was known. A deterministic polynomial-time algorithm was eventually found (running in time $\tilde{\mathcal{O}}(N^{12})$, later improved to $\tilde{\mathcal{O}}(N^6)$), but it is not used in practice, both because it is not particularly fast and because there is a better alternative.

1.1 Randomized Algorithms

The algorithm used in practice is *randomized*: the Miller–Rabin test decides primality with high probability in $\mathcal{O}(N^3)$ time. The idea is that if n is composite, then at least $3/4$ of the numbers in $\{2, \ldots, n-2\}$ are *witnesses* of the fact that n is composite, while a prime has no witness. Sampling one random number therefore detects a composite n with probability at least $3/4$, and after k independent samples a composite n goes undetected with probability at most $(1/4)^k$. If the test says "composite" it is always right; if it says "prime" it is wrong with probability at most $(1/4)^k$, which can be made as small as we like by increasing k.

2 Random Sampling

All that a randomized algorithm needs is the ability to *sample*. We assume that flipping a (fair) coin is an $\mathcal{O}(1)$ time primitive.

2.1 Sampling a Random Element of an Array

Consider an array with n slots (Figure 1) and assume for simplicity that n is a power of 2. Every index is a bit string of $\log_2 n = \log n$ bits, so we can sample a uniformly random element

by flipping one coin for each bit of the index. This takes $\mathcal{O}(\log n)$ time, and every index i is obtained with probability

$$\mathbb{P}(\text{index } i \text{ is chosen}) = \left(\frac{1}{2}\right)^{\log n} = \frac{1}{2^{\log n}} = \frac{1}{n}.$$

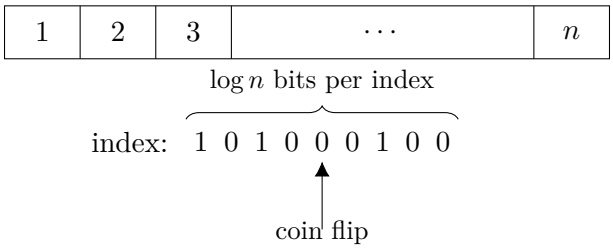


Figure 1: Sampling a random slot of an array with n slots: each of the $\log n$ bits of the index is the result of one coin flip.

We usually abstract away the $\mathcal{O}(\log n)$ and assume that an element can be sampled in constant time. This is only justified when the space we sample from has polynomial size (as a function of n), since then the number of coin flips is $\mathcal{O}(\log n)$; for a larger space the sampling time is no longer negligible.

2.2 With and Without Replacement

Random sampling can be done *with* or *without* replacement. We will almost always analyze sampling with replacement, because the analysis without replacement is more complex (the samples are no longer independent).

Suppose that we are looking for one particular element of the array. For every sample,

$$\mathbb{P}(\text{element} = \text{sample}) = \frac{1}{n}.$$

If we keep sampling (with replacement) until the element is found, then the expected number of samples, i.e., the number of steps of the algorithm, is

$$\mathbb{E}[\text{samples}] = \frac{1}{\mathbb{P}(\text{element} = \text{sample})} = n.$$

We will prove this in the next lecture, when we introduce the expectation and the geometric random variable. For now we can already compute the probability of *not* finding the element: a sample misses it with probability $1 - \frac{1}{n}$, and since the samples are independent,

$$\mathbb{P}(\text{not found after } k \text{ samples}) = \left(1 - \frac{1}{n}\right)^k, \qquad \mathbb{P}(\text{found within } k \text{ samples}) = 1 - \left(1 - \frac{1}{n}\right)^k.$$

3 Factoring

Factoring is the second motivating problem: given $n = pq$ with p, q prime, find p and q.

- Factoring is an *encryption primitive*: the security of widely used encryption schemes rests on it being hard. If factoring can be solved quickly, then the encryption is broken.
- No polynomial-time (in $N = \log n$) classical algorithm is known.
- There is a quantum algorithm that runs in polynomial time, *Shor's factoring algorithm* (time $\mathcal{O}(N^3)$).

- There is *no* known quantum algorithm that solves NP-complete problems in polynomial time.

Recall that NP is the class of problems whose solutions are easy to verify (in polynomial time), and that the NP-complete problems are the NP problems to which every NP problem reduces (equivalently, the problems that are reducible to 3-SAT and vice versa). Whether P = NP, i.e., whether NP-complete problems can be solved in polynomial time, is a famous open problem. A quantum computer can simulate a classical one with polynomial overhead, so if P = NP then quantum computers also solve NP-complete problems in polynomial time; conversely, if one could show that quantum computers cannot solve NP-complete problems in polynomial time, then P $\neq$ NP.

4 Search

The third problem, which we will study in detail, is *search*: we are given an (unsorted) array A with n slots, which can only be accessed through *queries* ("what is $A[i]$?"), and we want to find the index of a given element x. If the array were sorted, binary search would take $\mathcal{O}(\log n)$ queries; since it is not, linear search makes $\mathcal{O}(n)$ queries. Can we do better?

4.1 Lower Bound for Deterministic Algorithms

An upper bound of $\mathcal{O}(f(n))$ queries only needs one algorithm; a lower bound of $\Omega(f(n))$ has to hold for *all* algorithms. The standard tool for this is an *adversary argument*: given any algorithm, the adversary constructs an input on which the algorithm performs badly.

Theorem 1 ▶ Lower Bound for Deterministic Search

Every deterministic algorithm for search makes at least $n - 1$ queries in the worst case.

Proof. Fix a deterministic algorithm. The adversary does not commit to the input in advance; instead it answers the queries of the algorithm as they come. Whenever the algorithm queries a slot i, the adversary answers that $A[i] \neq x$ (and fills $A[i]$ with some other value).

Since the algorithm is deterministic, its kth query is determined by the answers to the first $k-1$ queries, so this defines one input for the algorithm. Suppose that the algorithm stops after $k \leqslant n-2$ queries and outputs an index j. At least $n-k \geqslant 2$ slots have not been queried, so the adversary can place x in an unqueried slot different from j. The resulting array is consistent with all the answers that were given, and the algorithm is wrong on it. Hence a correct algorithm makes at least $n-1$ queries. □

After $n-1$ queries that were all answered "no", only one slot is left and the algorithm knows that x is there without querying it, so the bound $n-1$ is tight.

4.2 Randomized and Quantum Algorithms

- The adversary argument breaks down for *randomized* algorithms: the queries are no longer determined by the previous answers, so the adversary cannot predict them. Nevertheless, randomized search needs $\Omega(n)$ expected queries. The proof uses *Yao's min-max principle*, which is a good general tool for proving lower bounds for randomized algorithms (next lecture).
- Grover's quantum algorithm solves search with $\mathcal{O}(\sqrt{n})$ queries (later in the course).

To show a *quantum advantage* (often called quantum supremacy in class) for a problem, we have to show that the problem needs $\Omega(f(n))$ queries classically but only $o(f(n))$ queries quantumly. Since randomized algorithms are the strongest classical algorithms, this means that we need lower bounds for *randomized* algorithms, not only for deterministic ones. This is why lower bounds for randomized algorithms play a bigger role in this course than lower bounds for deterministic algorithms.

5 Probability

We review the basic notions of (discrete) probability (Figure 2). A **sample space** S is a finite set of *simple events*, $|S| = n$. A **probability function** $\mathbb{P}\colon S \to [0,1] \subseteq \mathbb{R}$ assigns a probability to every simple event. An **event** is a subset $A \subseteq S$, with probability $\mathbb{P}(A) = \sum_{a \in A} \mathbb{P}(a)$. The probability function satisfies the *axioms* $\mathbb{P}(A) \geqslant 0$ for every event A, $\mathbb{P}(S) = 1$, and $\mathbb{P}(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} \mathbb{P}(A_i)$ for pairwise disjoint events $A_1, \ldots, A_n$ (i.e., $A_i \cap A_j = \emptyset$ for all $i \neq j$). A **random variable** is a function $X\colon S \to V$ from the sample space to a set of values V (usually $V \subseteq \mathbb{R}$), and for a value $r \in V$, $\mathbb{P}(X = r) = \sum_{s \in S\,:\,X(s)=r} \mathbb{P}(s)$.

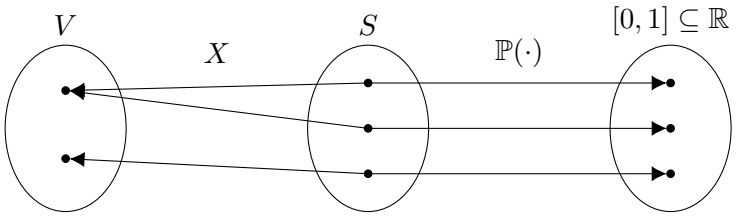


Figure 2: A sample space S of simple events. The probability function $\mathbb{P}(\cdot)$ maps every simple event to a number in $[0,1]$, and a random variable X maps every simple event to a value in V.

For example, in random sampling the sample space S is the set of the n possible outcomes of the $\log n$ coin flips, and the random variable X is the index that is sampled. Every $s \in S$ has $\mathbb{P}(s) = 1/n$, and $\mathbb{P}(X = i) = 1/n$ for every index i. In the next lecture we define the expectation $\mathbb{E}[X]$ of a random variable and prove that the expected number of samples until a given element is found is n.

Lecture Date: September 1, 2026

Lecture Topic: Query Complexity of Search, Coupon Collector Problem, Lower Bound for Randomized Algorithms, Chernoff Bounds

In this lecture, we study the query complexity of search for classical randomized algorithms. We analyze random sampling, relate it to the coupon collector problem, state a lower bound for randomized search algorithms using Yao's minmax principle, and introduce Chernoff bounds.

6 Query Complexity for Search

Recall the search problem of the last lecture (Figure 1): an array of n slots can only be accessed through *queries*, a query to a slot x returns the value $f(x)$, and the goal is to find the slot x with $f(x) = 1$. The cost of an algorithm is its *query complexity*, the number of queries that it makes.

Definition 2 ▶ Monte Carlo and Las Vegas Algorithms

A **Monte Carlo** algorithm is a randomized algorithm whose error probability is > 0.
A **Las Vegas** algorithm is a randomized algorithm whose error probability is $= 0$.

Random sampling *with replacement* queries in every step a slot chosen uniformly at random, so every query succeeds with probability $1/n$, and the number of queries until the first success is a *geometric* random variable with parameter $1/n$. Hence

$$\mathbb{E}[\text{queries until find}] = n,$$

which proves the claim of the last lecture. Stopped after a fixed number of queries, random sampling is a Monte Carlo algorithm, since it can fail to find the slot; run until the slot is found, it is a Las Vegas algorithm with n expected queries.

7 Coupon Collector Problem

Suppose that we want to distinguish 2 binary arrays (Figure 3). To do this with random sampling, all the n slots have to be accessed. The number of queries needed by random sampling (with replacement) to access all the n slots is the *coupon collector problem*. We show that

$$\mathbb{E}[\text{queries}] \leqslant \mathcal{O}(n \log n) \quad \text{and} \quad \mathbb{E}[\text{queries}] \geqslant \Omega(n \log n),$$

i.e., $\mathbb{E}[\text{queries}] = \Theta(n \log n)$.

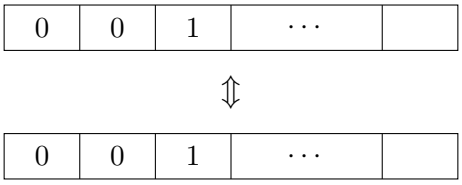


Figure 3: To distinguish 2 binary arrays.

Theorem 2 ▶ Coupon Collector

Let X be the number of queries made by random sampling (with replacement) until all the n slots are accessed. Then

$$\mathbb{E}[X] = nH_n = \Theta(n \log n),$$

where $H_n = \sum_{i=1}^{n} \frac{1}{i}$ is the nth harmonic number.

Proof. Write

$$X = X_1 + X_2 + \cdots + X_n,$$

where X_i is the number of queries made to access a new slot, after $i-1$ different slots have already been accessed. When $i-1$ slots have been accessed, a query accesses a new slot with probability $\frac{n-(i-1)}{n}$. Hence X_i is a geometric random variable,

$$X_i \sim G\left(\frac{n-(i-1)}{n}\right), \qquad \mathbb{E}[X_i] = \frac{n}{n-(i-1)}.$$

By linearity of expectation,

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}[X_1 + X_2 + \cdots + X_n] \\
&= \sum_{i=1}^{n} \mathbb{E}[X_i] \\
&= \sum_{i=1}^{n} \frac{n}{n-(i-1)} \\
&= n \sum_{i=1}^{n} \frac{1}{n-i+1} \\
&= nH_n.
\end{aligned}$$

Since $H_n = \Theta(\log n)$, we get $\mathbb{E}[X] = \Theta(n \log n)$. □

8 High Probability Bound

The following was given in class as an exercise. Its solution uses the Chernoff bounds, which are stated at the end of this lecture, so it is given in Section 11.

Exercise 1. Show that

$$\mathbb{P}(\text{any slot is accessed} \leqslant c \log n \text{ times}) \leqslant \frac{1}{n^2}.$$

The expectation of the number of queries is $\Theta(n \log n)$. We now show that $3n \log n$ queries are enough with high probability. Here log is the natural logarithm.

Theorem 3

If random sampling makes $3n \log n$ queries, then

$$\mathbb{P}(\text{fail}) \leqslant \frac{1}{n^2}, \qquad \text{i.e.,} \qquad \mathbb{P}(\text{Success}) \geqslant 1 - \frac{1}{n^2},$$

where "fail" means that some slot is not accessed.

Proof. Let B_i be the event that slot i is not accessed. A query does not access slot i with probability $1 - \frac{1}{n}$. Using $1 - x \leqslant e^{-x}$,

$$\begin{aligned}\mathbb{P}(B_i) = \mathbb{P}(\text{slot } i \text{ not accessed}) &\leqslant \left(1 - \frac{1}{n}\right)^{3n \log n} \\ &\leqslant e^{-\frac{1}{n} \cdot 3n \log n} \\ &= e^{-3 \ln n} \\ &= \frac{1}{n^3}.\end{aligned}$$

By the union bound,

$$\mathbb{P}\left(\bigcup_i B_i\right) \leqslant \sum_{i=1}^{n} \mathbb{P}(B_i) = \frac{n}{n^3} = \frac{1}{n^2}.$$

□

9 Lower Bound for Randomized Algorithms

Theorem 4

Any Las Vegas randomized algorithm for search needs $\Omega(n)$ expected queries.

The lower bound is shown using Yao's minmax principle.

Yao's Minmax Principle

Let A^{rand} be a randomized algorithm, and let $\mathbb{E}\left[A^{\text{rand}}\right]$ be its expected number of queries. Let μ be a probability distribution over the inputs, and let D_μ^{best} be the best deterministic algorithm for inputs that are chosen according to μ. Then

$$\mathbb{E}\left[A^{\text{rand}}\right] \geqslant \max_\mu \mathbb{E}_\mu\left[D_\mu^{\text{best}}\right] \geqslant \Omega(n).$$

The quantity $\mathbb{E}_\mu\left[D_\mu^{\text{best}}\right]$ is called the *distributional complexity*.

10 Chernoff Bounds

Let's say that a coin gives heads (H) with probability $\frac{1}{2} + \varepsilon$ and tails (T) with probability $\frac{1}{2} - \varepsilon$. If we toss the coin n times, then

$$\mathbb{E}[\text{Heads}] = n\left(\frac{1}{2} + \varepsilon\right) = \frac{n}{2} + n\varepsilon.$$

Chernoff bounds show that the number of heads is close to its expectation, so that

$$\mathbb{P}\left(\text{Heads} < \frac{n}{2}\right) \sim \text{very small}.$$

In general, let

$$X = \sum_{i=1}^{n} X_i, \qquad X_i = \begin{cases} 1 & \text{with probability } p_i, \\ 0 & \text{with probability } 1 - p_i, \end{cases}$$

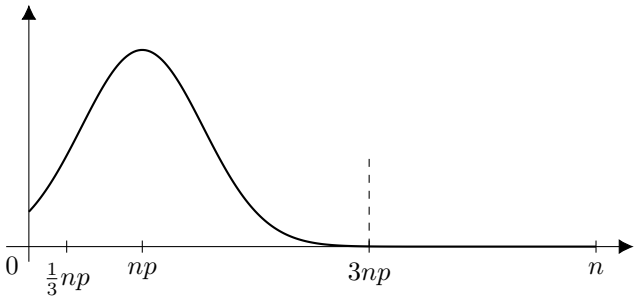


Figure 4: The distribution of X is concentrated around the mean np.

where the X_i are independent Bernoulli random variables. If $p_i = p$ for all i, then

$$\mathbb{E}[X] = np, \qquad \mathbb{V}[X] = np(1-p).$$

The distribution of X is concentrated around its mean np (Figure 4).

Theorem 5 ▶ Chernoff Bounds

Let $X = \sum_{i=1}^{n} X_i$ be as above and let $\mu = \mathbb{E}[X]$.

- Upper tail (UI): for $0 < \delta < 1$,

$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{3}}.$$

- For $R \geqslant 6\mu$,

$$\mathbb{P}(X \geqslant R) \leqslant \frac{1}{2^R}.$$

- Lower tail (LI): for $0 < \delta < 1$,

$$\mathbb{P}(X \leqslant (1-\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{2}}.$$

11 Solution of the Exercise

We now solve Exercise 1 with the Chernoff bounds of Theorem 5. The exercise says that the queries of random sampling are spread evenly over the slots: every slot is accessed about $\log n$ times, up to a constant c. We prove this in both directions. First, with $3n \log n$ queries no slot is accessed much *more* than the expected $3 \log n$ times (upper tail, $c = 18$). Second, with a constant factor more queries no slot is accessed much *less* than expected (lower tail, $c = 12$); we also explain why the lower tail needs the extra queries. As before, log is the natural logarithm.

Fix a slot i and suppose that random sampling makes T queries. Let

$$Y_i = \sum_{t=1}^{T} Z_t, \qquad Z_t = \begin{cases} 1 & \text{if the } t\text{th query accesses slot } i, \\ 0 & \text{otherwise,} \end{cases}$$

be the number of times that slot i is accessed. Every query accesses slot i with probability $1/n$, and the queries are independent (sampling with replacement), so the Z_t are independent Bernoulli random variables with $p_t = 1/n$, exactly as in Theorem 5, and

$$\mu = \mathbb{E}[Y_i] = \frac{T}{n}.$$

Lemma 6 ▶ Upper Tail: No Slot Is Accessed Too Often

If random sampling makes $3n \log n$ queries, then

$$\mathbb{P}(\text{some slot is accessed} \geqslant 18 \log n \text{ times}) \leqslant \frac{1}{n^2}.$$

Proof. Fix a slot i. Here $T = 3n \log n$, so $\mu = 3 \log n$, and $R = 18 \log n = 6\mu$. The second Chernoff bound of Theorem 5 gives

$$\mathbb{P}(Y_i \geqslant 18 \log n) \leqslant \frac{1}{2^{18 \log n}} = \frac{1}{n^{18 \log 2}} \leqslant \frac{1}{n^3},$$

where $2^{18 \log n} = e^{18 \log 2 \cdot \log n} = n^{18 \log 2}$ and $18 \log 2 > 12 > 3$. Let C_i be the event that slot i is accessed at least $18 \log n$ times. By the union bound,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} C_i\right) \leqslant \sum_{i=1}^{n} \mathbb{P}(C_i) \leqslant n \cdot \frac{1}{n^3} = \frac{1}{n^2}.$$ □

Lemma 7 ▶ Lower Tail: No Slot Is Accessed Too Rarely

If random sampling makes $24n \log n$ queries, then

$$\mathbb{P}(\text{some slot is accessed} \leqslant 12 \log n \text{ times}) \leqslant \frac{1}{n^2}.$$

Proof. Fix a slot i. Now $T = 24n \log n$, so $\mu = 24 \log n$. Take $\delta = 1/2$ in the lower tail bound of Theorem 5; then $(1 - \delta)\mu = 12 \log n$ and

$$\mathbb{P}(Y_i \leqslant 12 \log n) = \mathbb{P}(Y_i \leqslant (1 - \delta)\mu) \leqslant e^{-\frac{\mu \delta^2}{2}} = e^{-\frac{24 \log n}{8}} = e^{-3 \log n} = \frac{1}{n^3}.$$

Let D_i be the event that slot i is accessed at most $12 \log n$ times. By the union bound,

$$\mathbb{P}\left(\bigcup_{i=1}^{n} D_i\right) \leqslant \sum_{i=1}^{n} \mathbb{P}(D_i) \leqslant n \cdot \frac{1}{n^3} = \frac{1}{n^2}.$$ □

Why the lower tail needs more queries. With only $T = 3n \log n$ queries we have $\mu = 3 \log n$, and the lower tail bound gives, for every $0 < \delta < 1$,

$$\mathbb{P}(Y_i \leqslant (1 - \delta)\, 3 \log n) \leqslant e^{-\frac{3 \log n \cdot \delta^2}{2}} = \frac{1}{n^{3\delta^2/2}} > \frac{1}{n^{3/2}}.$$

After the union bound over the n slots this is more than $n^{-1/2}$, which is far from the n^{-2} that the exercise asks for. In general, with $T = a\, n \log n$ queries the same computation gives $n \cdot n^{-a\delta^2/2}$, which is at most $1/n^2$ as soon as $a \geqslant 6/\delta^2$; Lemma 7 is the choice $\delta = 1/2$, $a = 24$. The upper tail does not have this problem because the bound 2^{-R} for $R \geqslant 6\mu$ is much stronger.

Corollary 8 ▶ Every Slot Is Accessed $\Theta(\log n)$ Times

If random sampling makes $24n \log n$ queries, then with probability at least $1 - 2/n^2$ every slot is accessed more than $12 \log n$ and fewer than $144 \log n$ times.

Proof. The lower bound is Lemma 7. For the upper bound, repeat the proof of Lemma 6 with $\mu = 24 \log n$ and $R = 6\mu = 144 \log n$: $\mathbb{P}(Y_i \geqslant 144 \log n) \leqslant 2^{-144 \log n} \leqslant n^{-3}$, and the union bound gives $1/n^2$. The two failure events together have probability at most $2/n^2$. □

Lecture Date: September 3, 2026

Lecture Topic: Markov, Chebyshev and Chernoff Inequalities, Search, Birthday Paradox

In this lecture, we state and prove Markov's inequality and Chebyshev's inequality, and we state the Chernoff bound. We then look at the query complexity of search for classical and quantum algorithms, and we analyze the birthday paradox.

12 Markov's Inequality

Markov's inequality uses only the expectation $\mathbb{E}[X]$ of the random variable, which is also called the *first moment*.

Theorem 9 ► Markov's Inequality

Let $X \geqslant 0$ be a random variable. Then, for any $a > 0$,

$$\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}.$$

Equivalently, for any $b > 0$,

$$\mathbb{P}(X \geqslant b) \leqslant \frac{\mathbb{E}[X]}{b}.$$

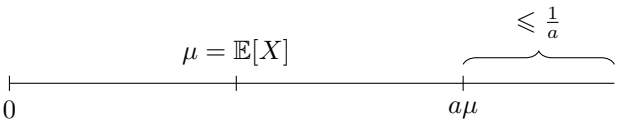


Figure 5: Markov's inequality: the probability that X is at least $a\mu$ is at most $1/a$.

Proof. Let

$$I = \begin{cases} 1 & \text{if } X \geqslant b, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$I \leqslant \frac{X}{b}.$$

This is true for $X \geqslant b$, since then $I = 1 \leqslant \frac{X}{b}$. It is also true for $X < b$, since then $I = 0$ and $X \geqslant 0$. Markov's inequality follows by taking expectations on both sides:

$$\mathbb{E}[I] \leqslant \mathbb{E}\left[\frac{X}{b}\right] = \frac{\mathbb{E}[X]}{b}.$$

Since

$$\mathbb{P}(I = 1) = \mathbb{P}(X \geqslant b) \qquad \text{and} \qquad \mathbb{P}(I = 0) = \mathbb{P}(X < b),$$

we have $\mathbb{E}[I] = \mathbb{P}(I = 1)$, and hence

$$\mathbb{P}(X \geqslant b) \leqslant \frac{\mathbb{E}[X]}{b}.$$

Taking $b = a\,\mathbb{E}[X]$ gives $\mathbb{P}(X \geqslant a\,\mathbb{E}[X]) \leqslant \frac{1}{a}$. □

13 Chebyshev's Inequality

Chebyshev's inequality uses the variance of the random variable, which is also called the *second moment*.

Definition 3 ▶ Variance and Standard Deviation

The **variance** of a random variable X is

$$\mathbb{V}[X] = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big].$$

The **standard deviation** of X is

$$\sigma = \sqrt{\mathbb{V}[X]}.$$

Theorem 10 ▶ Chebyshev's Inequality

For any random variable X and any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) \leqslant \frac{\mathbb{V}[X]}{a^2} = \frac{\sigma^2}{a^2}.$$

Here $|X - \mathbb{E}[X]|$ is the deviation of X from its mean.

Proof.

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) = \mathbb{P}\Big((X - \mathbb{E}[X])^2 \geqslant a^2\Big) \leqslant \frac{\mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]}{a^2} = \frac{\mathbb{V}[X]}{a^2},$$

where the inequality is Markov's inequality applied to the random variable $(X - \mathbb{E}[X])^2 \geqslant 0$. □

14 Chernoff Bound

The Chernoff bounds of the last lecture (Theorem 5) were restated in class with the independence assumption made explicit. For a sum $X = \sum_{i=1}^{n} X_i$ of *mutually independent* indicator random variables with $\mathbb{P}(X_i = 1) = p_i$, linearity of expectation gives $\mu = \mathbb{E}[X] = \sum_{i=1}^{n} p_i$, and for $0 < \delta < 1$

$$\mathbb{P}(X \geqslant (1+\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{3}}, \qquad \mathbb{P}(X < (1-\delta)\mu) \leqslant e^{-\frac{\mu\delta^2}{2}},$$

and $\mathbb{P}(X \geqslant R) \leqslant 2^{-R}$ for $R \geqslant 6\mu$.

15 Search

Let $f\colon D \to R$ be a function from a domain D to a range R (Figure 6). Given y, we ask: is there an x such that $f(x) = y$? We consider

$$f\colon [1, \ldots, N] \to [1, \ldots, N],$$

and we want to find an x such that $f(x) = k$. The function can be seen as an array with n slots, where n is the size of the domain, and the cost of an algorithm is the number of queries to f.

- Randomized classical algorithm: $\Theta(n)$ queries.
- Grover's algorithm: $\mathcal{O}(\sqrt{n})$ queries.

For the random sampling algorithm,

$$\mathbb{E}[\text{queries}] = \mathcal{O}(n), \qquad \mathbb{P}(\text{Success}) = \frac{1}{n}.$$

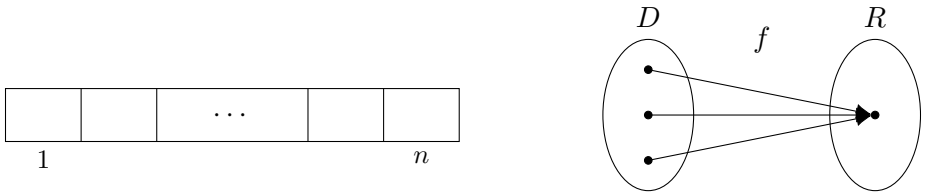


Figure 6: Left: an array with n slots. Right: a function $f\colon D \to R$.

Probability amplitude amplification. If the classical success probability is p, then in the quantum case it becomes $\sqrt{p}$.

16 Birthday Paradox

There are m people and n birthdays ($n = 365$). We are looking for a *collision*, i.e., $x \neq y$ such that

$$f(x) = f(y),$$

where $f(x)$ is the birthday of person x.

Let E_i be the event that the ith birthday is different from the first $i-1$ birthdays. Then

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{i=1}^{m} E_i\right) &= \mathbb{P}(\text{all birthdays different}) \\
&= \mathbb{P}(E_1) \cdot \mathbb{P}(E_2 \mid E_1) \cdot \mathbb{P}(E_3 \mid E_2 \cap E_1) \cdots \\
&= 1 \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{m-1}{n}\right) \\
&= \prod_{i=0}^{m-1} \left(1 - \frac{i}{n}\right) \\
&\leqslant \prod_{i=0}^{m-1} e^{-i/n} \\
&= e^{-\sum_{i=0}^{m-1} \frac{i}{n}} \\
&= e^{-\frac{(m-1)m}{2n}}.
\end{aligned}$$

For $m = \sqrt{2n} + 1$,

$$\mathbb{P}(\text{all birthdays different}) \leqslant e^{-\frac{\sqrt{2n}(\sqrt{2n}+1)}{2n}} \leqslant e^{-\frac{2n}{2n}} = \frac{1}{e} \leqslant \frac{1}{2}.$$

Hence $\mathcal{O}(\sqrt{n})$ people are enough to have a constant probability of collision.

16.1 Expected Number of Collisions

The birthday paradox can be seen as throwing m balls into n bins (Figure 7). We show that for $m \approx \sqrt{2n}$,

$$\mathbb{E}[\#\text{ collisions}] \geqslant 1.$$

Let X be the number of collisions. X takes values from 0 to $\binom{m}{2}$. Write

$$X = \sum_{i \neq j} X_{ij}, \qquad X_{ij} = \begin{cases} 1 & \text{if } f(i) = f(j), \\ 0 & \text{else,} \end{cases}$$

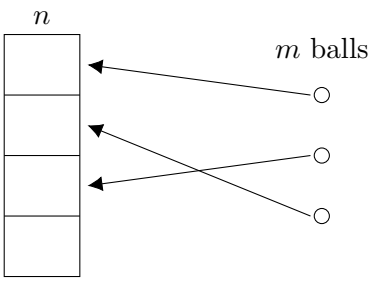


Figure 7: m balls are thrown into n bins.

where the sum is over the $\binom{m}{2}$ pairs $i \neq j$. The random variables X_{ij} are *pairwise independent*. By linearity of expectation,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i\neq j} \mathbb{E}[X_{ij}] \\ &= \sum_{i\neq j} \mathbb{P}(X_{ij} = 1) \\ &= \sum_{i\neq j} \frac{1}{n} \\ &= \binom{m}{2}\frac{1}{n} = \frac{m(m-1)}{2n} \sim \frac{m^2}{n}.\end{aligned}$$

Exercise 2. Show that

$$\mathbb{P}(X \geqslant 1) \geqslant \Theta(1),$$

i.e., the birthday paradox occurs.

The hint given in class is to compute the variance

$$\mathbb{V}[X] = \mathbb{V}\left[\sum_{i\neq j} X_{ij}\right]$$

and to use Chebyshev's inequality (Theorem 10): if the variance of X is small compared to $\mathbb{E}[X]^2$, then X is rarely far from its mean, and in particular rarely 0. We first prove the claim that the X_{ij} are pairwise independent, which is what makes the variance easy to compute.

Lemma 11 ▶ Variance of the Number of Collisions

For any two different pairs $\{i, j\} \neq \{k, l\}$,

$$\mathbb{E}[X_{ij}X_{kl}] = \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}] = \frac{1}{n^2},$$

i.e., X_{ij} and X_{kl} are independent. Consequently

$$\mathbb{V}[X] = \sum_{i\neq j} \mathbb{V}[X_{ij}] = \binom{m}{2}\frac{1}{n}\left(1 - \frac{1}{n}\right) \leqslant \mathbb{E}[X].$$

Proof. Since $X_{ij}X_{kl} = 1$ exactly when both indicators are 1,

$$\mathbb{E}[X_{ij}X_{kl}] = \mathbb{P}(f(i) = f(j) \text{ and } f(k) = f(l)).$$

There are two cases.

- The pairs are disjoint, $\{i,j\} \cap \{k,l\} = \emptyset$. Then the two events depend on the birthdays of four different people, which are independent, so the probability is $\mathbb{P}(f(i) = f(j)) \cdot \mathbb{P}(f(k) = f(l)) = 1/n^2$.
- The pairs share one person, say $\{i,j\}$ and $\{i,k\}$ with $j \neq k$. The birthdays $f(j)$ and $f(k)$ are independent of $f(i)$ and of each other, and each equals a given day b with probability $1/n$, so

$$\mathbb{P}(f(j) = f(i) \text{ and } f(k) = f(i)) = \sum_{b=1}^{n} \mathbb{P}(f(i) = b) \cdot \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n^2}.$$

In both cases $\mathbb{E}[X_{ij}X_{kl}] = 1/n^2 = \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]$. For two indicator random variables this is the same as independence, since their joint distribution is determined by $\mathbb{P}(X_{ij} = 1)$, $\mathbb{P}(X_{kl} = 1)$ and $\mathbb{P}(X_{ij} = X_{kl} = 1)$.

For the variance, write $X - \mathbb{E}[X] = \sum_{i \neq j}(X_{ij} - \mathbb{E}[X_{ij}])$ and expand the square:

$$\begin{aligned}
\mathbb{V}[X] &= \mathbb{E}\left[\Big(\sum_{i\neq j}(X_{ij} - \mathbb{E}[X_{ij}])\Big)^2\right] \\
&= \sum_{i\neq j} \mathbb{E}\Big[(X_{ij} - \mathbb{E}[X_{ij}])^2\Big] + \sum_{\{i,j\}\neq\{k,l\}} \mathbb{E}[(X_{ij} - \mathbb{E}[X_{ij}])(X_{kl} - \mathbb{E}[X_{kl}])] \\
&= \sum_{i\neq j} \mathbb{V}[X_{ij}] + \sum_{\{i,j\}\neq\{k,l\}} (\mathbb{E}[X_{ij}X_{kl}] - \mathbb{E}[X_{ij}]\,\mathbb{E}[X_{kl}]) = \sum_{i\neq j} \mathbb{V}[X_{ij}],
\end{aligned}$$

where the second sum vanishes by the first part. Finally X_{ij} is an indicator with $\mathbb{P}(X_{ij} = 1) = 1/n$, so $X_{ij}^2 = X_{ij}$ and

$$\mathbb{V}[X_{ij}] = \mathbb{E}\Big[X_{ij}^2\Big] - \mathbb{E}[X_{ij}]^2 = \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n}\left(1 - \frac{1}{n}\right) \leqslant \frac{1}{n}.$$

Summing over the $\binom{m}{2}$ pairs gives $\mathbb{V}[X] \leqslant \binom{m}{2}\frac{1}{n} = \mathbb{E}[X]$. □

Theorem 12 ▶ The Birthday Paradox Occurs

If $m \geqslant 2\sqrt{n} + 1$, then

$$\mathbb{P}(X \geqslant 1) \geqslant \frac{1}{2}.$$

In words: with $\mathcal{O}(\sqrt{n})$ people there is a collision with constant probability.

Proof. The event $X = 0$ (no collision) is contained in the event $|X - \mathbb{E}[X]| \geqslant \mathbb{E}[X]$, since $X = 0$ means $X - \mathbb{E}[X] = -\mathbb{E}[X]$. By Chebyshev's inequality (Theorem 10) with $a = \mathbb{E}[X]$, and then Lemma 11,

$$\mathbb{P}(X = 0) \leqslant \mathbb{P}(|X - \mathbb{E}[X]| \geqslant \mathbb{E}[X]) \leqslant \frac{\mathbb{V}[X]}{\mathbb{E}[X]^2} \leqslant \frac{\mathbb{E}[X]}{\mathbb{E}[X]^2} = \frac{1}{\mathbb{E}[X]}.$$

For $m \geqslant 2\sqrt{n} + 1$, since $m(m-1)$ grows with m,

$$\mathbb{E}[X] = \frac{m(m-1)}{2n} \geqslant \frac{(2\sqrt{n}+1)\cdot 2\sqrt{n}}{2n} = \frac{4n + 2\sqrt{n}}{2n} \geqslant 2,$$

so $\mathbb{P}(X = 0) \leqslant 1/2$, and $\mathbb{P}(X \geqslant 1) = 1 - \mathbb{P}(X = 0) \geqslant 1/2$. □

The direct computation of $\mathbb{P}(\text{all birthdays different})$ above already gave $\mathbb{P}(X = 0) \leqslant 1/e$ for $m = \sqrt{2n} + 1$. The point of this *second moment* argument is that it only uses $\mathbb{E}[X]$ and $\mathbb{V}[X]$, so it also works when the exact probability cannot be computed.

17 Quantum Birthday Paradox

The birthday paradox, seen as an algorithm, solves the *collision problem*: given query access to a function f, find $x \neq y$ with $f(x) = f(y)$. Querying $m = \sqrt{2n} + 1$ random points finds a collision with probability at least $1/2$, so $\mathcal{O}(\sqrt{n})$ queries suffice classically. The quantum version of the problem is studied later in the course; here we show that classically $\sqrt{n}$ queries are also necessary. An algorithm makes k queries

$$f(x_1), \ldots, f(x_k),$$

where the point x_s may depend on the earlier answers $f(x_1), \ldots, f(x_{s-1})$ and on random coins.

Exercise 3. Show a $\sqrt{n}$ lower bound in the classical case.

We use the same model as in the birthday paradox: $f\colon D \to [n]$, where $[n] = \{1, \ldots, n\}$, is a *random function*, i.e., every value $f(x)$ is uniform in $[n]$ and independent of all the other values (the birthday of person x). The algorithm *finds a collision* if it outputs $x \neq y$ with $f(x) = f(y)$. We may assume that the algorithm queries the two points that it outputs: if it does not, it can do so with at most 2 extra queries, which does not change the bound below. We may also assume that the k queried points are different, since repeating a query gives no new information.

Theorem 13 ▶ Classical Lower Bound for Collision Finding

Let $f\colon D \to [n]$ be a random function. Every classical algorithm (deterministic or randomized, adaptive) that makes k queries finds a collision with probability at most

$$\frac{\binom{k}{2}}{n} = \frac{k(k-1)}{2n}.$$

In particular, to find a collision with probability at least $1/2$ it needs $k > \sqrt{n}$ queries, so the classical query complexity of the collision problem is $\Omega(\sqrt{n})$.

Proof. Since the algorithm queries the points that it outputs, it can only find a collision if two of its queried values agree. Let E_s be the event that the sth query collides with an earlier one, i.e., $f(x_s) \in \{f(x_1), \ldots, f(x_{s-1})\}$. A collision among the k queries means that some E_s occurs, so by the union bound

$$\mathbb{P}(\text{collision}) \leqslant \sum_{s=1}^{k} \mathbb{P}(E_s).$$

We claim that $\mathbb{P}(E_s) \leqslant \frac{s-1}{n}$. Consider the moment just before the sth query. The algorithm chooses x_s from the $s-1$ answers seen so far and from its random coins; condition on all of these. Then x_s is a fixed point that has not been queried before, so its value $f(x_s)$ is still a uniformly random element of $[n]$, independent of the $s-1$ values $f(x_1), \ldots, f(x_{s-1})$, which are now fixed. The probability that $f(x_s)$ equals one of these $s-1$ fixed values is therefore at most $\frac{s-1}{n}$, no matter how the algorithm chose x_s. Since this holds for every outcome of the first $s-1$ steps, it also holds without conditioning, which proves the claim. Hence

$$\mathbb{P}(\text{collision}) \leqslant \sum_{s=1}^{k} \frac{s-1}{n} = \frac{0 + 1 + \cdots + (k-1)}{n} = \frac{k(k-1)}{2n}.$$

If the algorithm succeeds with probability at least $1/2$, then $\frac{k(k-1)}{2n} \geqslant \frac{1}{2}$, i.e., $k^2 > k(k-1) \geqslant n$, so $k > \sqrt{n}$. □

Worst case inputs. Theorem 13 is a statement about a *random* input f: the distributional complexity of the collision problem under the uniform distribution on functions is $\Omega(\sqrt{n})$. By Yao's minmax principle from the lecture of September 1 (in its version for algorithms that are allowed to err), this is also a lower bound for every randomized algorithm on its worst case input. Together with the birthday paradox, which finds a collision with $\mathcal{O}(\sqrt{n})$ queries, the classical query complexity of finding a collision is $\Theta(\sqrt{n})$.

Lecture Date: September 24, 2026

Lecture Topic: Deutsch's Algorithm, Phase Flip, Walsh–Hadamard Transform, Deutsch–Jozsa Algorithm

In this lecture, we see the first quantum algorithms that beat every classical algorithm in the query model. We study Deutsch's algorithm, which decides with a single query whether a function $f\colon \{0,1\} \to \{0,1\}$ is constant or balanced, the *phase flip* form of a quantum query, the n-qubit Hadamard (Walsh–Hadamard) transform, and the Deutsch–Jozsa algorithm, which does the same for $f\colon \{0,1\}^n \to \{0,1\}$, again with a single query.

18 Deutsch's Algorithm

Problem 1 ▶ Deutsch's Problem

Given a function $f\colon \{0,1\} \to \{0,1\}$ as a black box, determine whether f is *constant* ($f(0) = f(1)$) or *balanced* ($f(0) \neq f(1)$).

Classically we need 2 queries: after one query, say $f(0)$, both answers are still possible. Quantumly, Deutsch's algorithm does it with 1 query.

18.1 The Quantum Query U_f

A quantum circuit is reversible, but f need not be (a constant f forgets its input). We therefore convert f into a reversible *quantum version* U_f: a controlled-f gate modeled on the CNOT gate (Figure 8). CNOT maps $|a\rangle|b\rangle \mapsto |a\rangle|a \oplus b\rangle$, and U_f maps

$$U_f\colon |a\rangle|b\rangle \longmapsto |a\rangle|b \oplus f(a)\rangle.$$

The first qubit (the input a) is left unchanged, and $f(a)$ is XORed into the second qubit. Applying U_f twice gives back $|a\rangle|b\rangle$, so U_f is reversible. One application of U_f counts as one query to f.

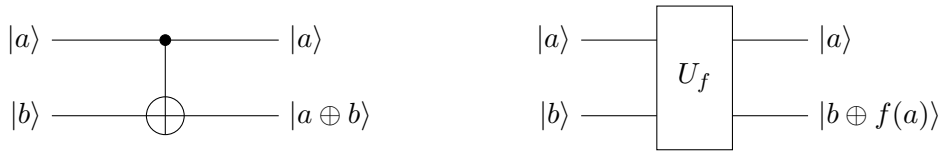


Figure 8: Left: the CNOT gate. Right: the quantum query U_f, a controlled-f gate.

18.2 The Circuit

Deutsch's algorithm is usually given in the form of Figure 9: the first qubit starts in $|0\rangle$ and the second in $|1\rangle$; both go through a Hadamard gate, then through U_f, then through a Hadamard gate again, and are measured. We will show that the measurement of the first qubit gives $|0\rangle$ if f is constant and $|1\rangle$ if f is balanced (the second qubit always gives $|1\rangle$).

Recall that the Hadamard gate H acts as

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

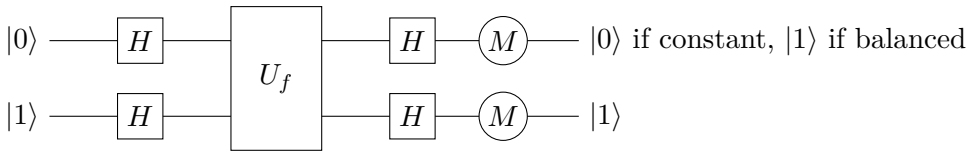


Figure 9: Deutsch's algorithm. It makes one query (U_f); the first measurement decides whether f is constant or balanced.

or, in one formula, for $a \in \{0,1\}$,

$$H|a\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^a|1\rangle) = \frac{1}{\sqrt{2}} \sum_{b\in\{0,1\}} (-1)^{ab}|b\rangle. \tag{1}$$

Since $H^2 = I$, applying H to the right-hand side of Equation (1) gives back $|a\rangle$.

Theorem 14 ▶ Deutsch's Algorithm

The circuit of Figure 9 makes one query to f, and measuring its first qubit gives $f(0) \oplus f(1)$ with probability 1. Hence the outcome is 0 if f is constant and 1 if f is balanced.

Proof. After the first two Hadamard gates the state is

$$\begin{aligned} |\psi_1\rangle = H|0\rangle \otimes H|1\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= \frac{1}{2}(|0\rangle(|0\rangle - |1\rangle) + |1\rangle(|0\rangle - |1\rangle)). \end{aligned}$$

Applying U_f to each term, and using that $|0 \oplus c\rangle - |1 \oplus c\rangle = (-1)^c(|0\rangle - |1\rangle)$ for $c \in \{0,1\}$ (for $c = 1$ the two kets are swapped),

$$\begin{aligned} |\psi_2\rangle = U_f|\psi_1\rangle &= \frac{1}{2}(|0\rangle(|0 \oplus f(0)\rangle - |1 \oplus f(0)\rangle) + |1\rangle(|0 \oplus f(1)\rangle - |1 \oplus f(1)\rangle)) \\ &= \frac{1}{2}\Big(|0\rangle(-1)^{f(0)}(|0\rangle - |1\rangle) + |1\rangle(-1)^{f(1)}(|0\rangle - |1\rangle)\Big) \\ &= \frac{1}{\sqrt{2}}\Big((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\Big) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\ &= (-1)^{f(0)}\frac{1}{\sqrt{2}}\Big(|0\rangle + (-1)^{f(0)\oplus f(1)}|1\rangle\Big) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \end{aligned}$$

where in the last step we used $(-1)^{f(1)} = (-1)^{f(0)}(-1)^{f(0)\oplus f(1)}$. The first qubit is now exactly the right-hand side of Equation (1) with $a = f(0) \oplus f(1)$, up to the sign $(-1)^{f(0)}$. Applying H to it therefore gives

$$|\psi_3\rangle = (-1)^{f(0)}|f(0) \oplus f(1)\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

The factor $(-1)^{f(0)}$ is a global phase and does not affect the measurement, so measuring the first qubit gives $f(0) \oplus f(1)$ with probability 1. (Applying H to the second qubit gives $|1\rangle$, which is why the second measurement in Figure 9 always gives 1.) □

19 Phase Flip

The computation above shows a general trick. If the second (target) qubit of U_f is in the state $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, then for every input x

$$U_f\,|x\rangle\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = (-1)^{f(x)}\,|x\rangle\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

i.e., the target qubit is unchanged and the value $f(x)$ shows up as a sign in front of $|x\rangle$. Since the target qubit never changes, we drop it and write the query in the short-hand *phase flip* form

$$U_f\colon |x\rangle \longmapsto (-1)^{f(x)}|x\rangle,$$

instead of the original form $U_f\colon |a\rangle|b\rangle \mapsto |a\rangle|b \oplus f(a)\rangle$. By linearity, U_f maps a superposition $\sum_x \alpha_x|x\rangle$ to $\sum_x \alpha_x(-1)^{f(x)}|x\rangle$: one query flips the sign of every $|x\rangle$ with $f(x) = 1$ at once.

20 The n-Qubit Hadamard Transform

To generalize Deutsch's algorithm to n bits we need the Hadamard gate applied to each of n qubits. Start with $n = 2$ and let $|x\rangle = |x_1x_2\rangle = |x_1\rangle|x_2\rangle$. Using Equation (1) on each qubit,

$$\begin{aligned}(H \otimes H)|x\rangle &= H|x_1\rangle \otimes H|x_2\rangle \\ &= \left(\frac{1}{\sqrt{2}}\sum_{y_1\in\{0,1\}}(-1)^{x_1y_1}|y_1\rangle\right)\left(\frac{1}{\sqrt{2}}\sum_{y_2\in\{0,1\}}(-1)^{x_2y_2}|y_2\rangle\right) \\ &= \frac{1}{2}\sum_{y\in\{0,1\}^2}(-1)^{x_1y_1+x_2y_2}|y\rangle = \frac{1}{2}\sum_{y\in\{0,1\}^2}(-1)^{x\cdot y}|y\rangle,\end{aligned}$$

where $x \cdot y = x_1y_1 \oplus x_2y_2$ is the inner product of the bit strings modulo 2 (only the parity of the exponent matters). In general, for $x, y \in \{0,1\}^n$ we write

$$x \cdot y = x_1y_1 \oplus x_2y_2 \oplus \cdots \oplus x_ny_n.$$

Lemma 15 ▶ Walsh–Hadamard Transform

The n-qubit Hadamard transform $H^{\otimes n} = H \otimes H \otimes \cdots \otimes H$ (the *Walsh–Hadamard transform*) acts on a basis state $|x\rangle = |x_1x_2\cdots x_n\rangle$ as

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}}\sum_{y\in\{0,1\}^n}(-1)^{x\cdot y}|y\rangle.$$

Proof. Exactly as for $n = 2$: apply Equation (1) to each of the n qubits and multiply out the n sums. The exponent is $x_1y_1 + \cdots + x_ny_n$, and (-1) to this power only depends on its parity $x \cdot y$. □

The most important special case is $x = 0^n = 00\cdots0$: then $x \cdot y = 0$ for every y, and

$$H^{\otimes n}|00\cdots0\rangle = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle$$

is the uniform superposition of all 2^n bit strings.

21 Deutsch–Jozsa Algorithm

Problem 2 ▶ Deutsch–Jozsa Problem

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, with the *promise* that f is either constant or balanced (i.e., $f(x) = 0$ for exactly half of the 2^n inputs), determine which is the case.

This is the same problem as Deutsch's, but on n bits. It is a *promise problem*: we are told in advance that f is one of the two kinds. Let $N = 2^n$ be the number of inputs.

21.1 Classical Query Complexity

- *Deterministic:* in the worst case $2^{n-1}+1 = N/2+1$ queries are needed. If an algorithm has seen only $N/2$ values and they are all equal, then f may still be either constant or balanced. The same holds for a randomized algorithm that has to succeed with probability 1: in the worst case it needs $2^{n-1}+1$ queries.
- *Randomized (Monte Carlo):* query f on α independent uniformly random inputs. If two of the answers differ, output "balanced"; otherwise output "constant". If f is constant, we are always right. If f is balanced, every answer is 0 or 1 with probability $1/2$ each, so all α answers agree, and we fail, with probability

$$2 \cdot \left(\frac{1}{2}\right)^{\alpha} = \left(\frac{1}{2}\right)^{\alpha-1}.$$

  A constant number of queries gives a constant error probability, and $\alpha = \mathcal{O}(\log N)$ queries bring the error probability down to $1/N$.
- *Quantum:* the Deutsch–Jozsa algorithm needs 1 query (and n measurements) and makes no error.

21.2 The Algorithm

The circuit (Figure 10) is Deutsch's circuit with n input qubits: n qubits in $|0\rangle$ and one qubit in $|1\rangle$ go through Hadamard gates, then through U_f (one query), then the n input qubits go through Hadamard gates again and are measured. If any of the n measurements gives 1, then f is balanced; if they all give 0, then f is constant.

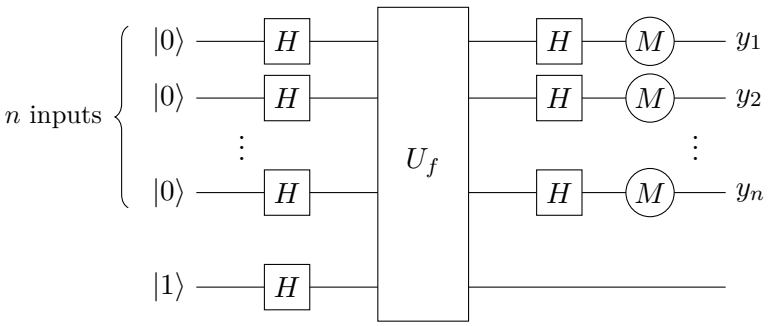


Figure 10: The Deutsch–Jozsa algorithm. It makes one query and n measurements; the last qubit is not needed after the query and is not measured.

Theorem 16 ▶ Deutsch–Jozsa Algorithm

In the circuit of Figure 10, measuring the n input qubits gives $y = 0^n$ with probability 1 if f is constant, and with probability 0 if f is balanced. Hence, if any measurement gives 1, then f is balanced; otherwise f is constant.

Proof. We use the phase flip form of U_f: the last qubit is $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ throughout and is dropped. The n input qubits start in $|00\cdots0\rangle$. After the first layer of Hadamard gates,

$$H^{\otimes n}|00\cdots0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle.$$

The query U_f flips the sign of every $|x\rangle$ with $f(x) = 1$:

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle.$$

After the second layer of Hadamard gates, by Lemma 15,

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle = \frac{1}{2^n} \sum_{y \in \{0,1\}^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)+x \cdot y} |y\rangle.$$

Look at the amplitude of $|y\rangle$ for $y = 0^n$. Since $x \cdot 0^n = 0$ for every x, it is

$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} = \begin{cases} \pm 1 & \text{if } f \text{ is constant (all } 2^n \text{ terms are equal),} \\ 0 & \text{if } f \text{ is balanced (half the terms are } +1\text{, half are } -1\text{).} \end{cases}$$

If f is constant, the state is $\pm|00\cdots0\rangle$ (all other amplitudes must be 0, since the amplitudes squared sum to 1), so the measurement gives 0^n with probability 1. If f is balanced, the amplitude of $|00\cdots0\rangle$ is 0, so the measurement gives 0^n with probability 0, i.e., at least one of the n measurements gives 1. □

For $n = 1$ this is exactly Deutsch's algorithm. In both cases a single query suffices, whereas any classical algorithm with zero error needs $2^{n-1} + 1$ queries.

Lecture Date: September 29, 2026

Lecture Topic: Walsh–Hadamard Transform Recap, Bernstein–Vazirani Algorithm, Simon's Algorithm

In this lecture, we first recall the quantum query U_f, the Walsh–Hadamard transform and the analysis of the Deutsch–Jozsa algorithm from the last lecture. We then see two more problems that a quantum algorithm solves with far fewer queries than any classical algorithm: the *Bernstein–Vazirani* problem (one quantum query instead of n classical queries) and *Simon's problem* ($\mathcal{O}(n)$ quantum queries instead of $\Omega\left(\sqrt{2^n}\right)$ classical queries). Simon's algorithm is the first one where the quantum speed-up is exponential.

22 Recap: Quantum Queries and the Walsh–Hadamard Transform

We recall the tools of the last lecture that both algorithms below use; the proofs are in the scribe of September 24. A function $f\colon \{0,1\}^n \to \{0,1\}^m$ is given as a black box, and in the quantum query model the black box is the unitary

$$U_f\colon |x\rangle|b\rangle \longmapsto |x\rangle|b \oplus f(x)\rangle,$$

which XORs $f(x)$ into the *target* register and leaves the *control* register $|x\rangle$ unchanged (Figure 8). Every classical circuit can be made into a reversible quantum circuit, so whenever f has a classical circuit, U_f has a quantum circuit of about the same size. One application of U_f is one query. The Hadamard gate acts as $H|a\rangle = \frac{1}{\sqrt{2}}\sum_{y\in\{0,1\}}(-1)^{ay}|y\rangle$, and on n qubits, with $x \cdot y = x_1y_1 \oplus \cdots \oplus x_ny_n$ the inner product of bit strings modulo 2, the *Walsh–Hadamard transform* (Lemma 15) is

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle, \qquad \text{in particular} \qquad H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} |y\rangle.$$

It is the *Fourier transform* over the group $\{0,1\}^n$ (with $\oplus$ as the group operation): the sign $(-1)^{x\cdot y}$ plays the role that the root of unity $e^{2\pi i\, xy/N}$ plays in the usual discrete Fourier transform. We will use the following sum many times; it is the Fourier-transform fact that "a nonzero character sums to zero".

Lemma 17 ▶ Sum of Signs

For every $z \in \{0,1\}^n$,

$$\sum_{x\in\{0,1\}^n} (-1)^{z\cdot x} = \begin{cases} 2^n & \text{if } z = 0^n, \\ 0 & \text{if } z \neq 0^n. \end{cases}$$

Proof. If $z = 0^n$, every term is $(-1)^0 = 1$ and there are 2^n terms. If $z \neq 0^n$, pick an index i with $z_i = 1$ and let e_i be the string with a single 1 in position i. The map $x \mapsto x \oplus e_i$ pairs every x with a different string (applying it twice gives x back), so it splits $\{0,1\}^n$ into 2^{n-1} disjoint pairs $\{x, x \oplus e_i\}$. The inner product is linear modulo 2, so $z\cdot(x\oplus e_i) = z\cdot x \oplus z\cdot e_i = z\cdot x \oplus z_i = z\cdot x \oplus 1$. So the two strings of every pair contribute $(-1)^{z\cdot x} + (-1)^{z\cdot x\oplus 1} = 0$, and the whole sum is 0. □

In the Deutsch–Jozsa circuit (Figure 10) the n control qubits start in $|0^n\rangle$ and the target qubit in $|1\rangle$. After the first layer of Hadamard gates the target is $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, so the query acts

on the control register as the *phase flip* $|k\rangle \mapsto (-1)^{f(k)}|k\rangle$ and the target never changes again. After the second layer of Hadamard gates the control register is in the state

$$\sum_{y\in\{0,1\}^n} \alpha_y |y\rangle, \qquad \alpha_y = \frac{1}{2^n} \sum_{k\in\{0,1\}^n} (-1)^{f(k)+k\cdot y}, \tag{2}$$

and $|\alpha_{0^n}|^2 = \left|\frac{1}{2^n}\sum_k (-1)^{f(k)}\right|^2$ is 1 if f is constant and 0 if f is balanced, which is Theorem 16. This computation was redone on the board; Equation (2) is all that the next section needs.

23 Bernstein–Vazirani Algorithm

Problem 3 ▶ Bernstein–Vazirani Problem

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, with the *promise* that

$$f(x) = s \cdot x = s_1x_1 \oplus s_2x_2 \oplus \cdots \oplus s_nx_n$$

for some unknown n-bit string s, find s.

Again this is a promise problem: we are told that f has this form. Classically, n queries suffice and are needed: querying the unit string e_i (a single 1 in position i) returns $f(e_i) = s_i$, one bit of s per query.

23.1 The Algorithm

The quantum algorithm uses exactly the Deutsch–Jozsa circuit (Figure 10), with the same analysis. It makes one query.

Theorem 18 ▶ Bernstein–Vazirani Algorithm

Run the Deutsch–Jozsa circuit on $f(x) = s \cdot x$. Then measuring the n control qubits gives $y = s$ with probability 1. Hence s is found with a single query.

Proof. By Equation (2) (with the summation index k renamed x), before the measurement the control register is in the state $\sum_y \alpha_y|y\rangle$ with

$$\alpha_y = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{s\cdot x \oplus y\cdot x} = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{(s\oplus y)\cdot x},$$

where we used that only the parity of the exponent matters and that $s\cdot x \oplus y\cdot x = (s\oplus y)\cdot x$ (the inner product is bilinear over $\{0,1\}$, i.e., modulo 2). By Lemma 17 applied with $z = s \oplus y$,

$$\alpha_y = \begin{cases} 1 & \text{if } y = s, \\ 0 & \text{if } y \neq s. \end{cases}$$

So the state is exactly $|s\rangle$, and the measurement gives s with probability 1. □

The Deutsch–Jozsa algorithm and the Bernstein–Vazirani algorithm are the same circuit; the difference is only in what the promise says about f and in how we read the measured string y.

Query complexity versus running time. The query model only counts the calls to U_f. The running time of the algorithm in the *quantum Turing machine* (or quantum circuit) model counts all the gates, and the two measures are not always the same: an algorithm can make few queries but need many other gates. For Deutsch–Jozsa and Bernstein–Vazirani the circuit has, besides the single query, only $2n + 1$ Hadamard gates and n measurements, so here both measures are small.

24 Simon's Algorithm

Problem 4 ▶ Simon's Problem

Given $f\colon \{0,1\}^n \to \{0,1\}^n$ as a black box, with the *promise* that there is a string $s \in \{0,1\}^n$ such that for all x, y,

$$f(x) = f(y) \iff x = y \text{ or } x = y \oplus s \qquad \text{(equivalently } s = x \oplus y\text{)},$$

find s.

If $s \neq 0^n$, then f is a 2-*to*-1 function: the inputs come in pairs $\{x, x \oplus s\}$, and the two inputs of a pair map to the same output while different pairs map to different outputs (Figure 11). If $s = 0^n$, then $f(x) = f(y)$ only for $x = y$, i.e., f is 1-*to*-1. So the problem contains the decision problem "is f 1-to-1 or 2-to-1?" and, in the second case, asks for the period s of f. We write $N = 2^n$.

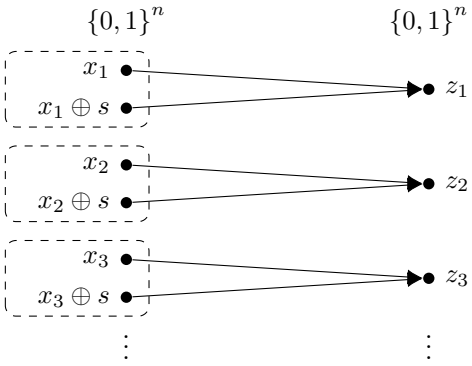


Figure 11: Simon's promise for $s \neq 0^n$: f is 2-to-1, and the two inputs of every pair $\{x, x \oplus s\}$ map to the same output. The range of f has 2^{n-1} elements.

Query complexity. Classically, Simon's problem needs $\Omega\left(\sqrt{N}\right) = \Omega\left(\sqrt{2^n}\right)$ queries, even for randomized algorithms (next lecture). Simon's quantum algorithm uses $\mathcal{O}(n)$ queries. The only expensive classical part of the algorithm is a Gaussian elimination on an $n \times n$ system of linear equations over $\{0,1\}$, which takes $\mathcal{O}(n^3)$ time. This is an exponential speed-up.

24.1 The Circuit

The circuit (Figure 12) has two registers of n qubits each, all starting in $|0\rangle$. The control register goes through Hadamard gates, then both registers go through U_f (one query), then the control register goes through Hadamard gates again and is measured, giving a string $y \in \{0,1\}^n$. Since f has n output bits, we use U_f in its original form $|x\rangle|b\rangle \mapsto |x\rangle|b \oplus f(x)\rangle$ and not the phase flip form.

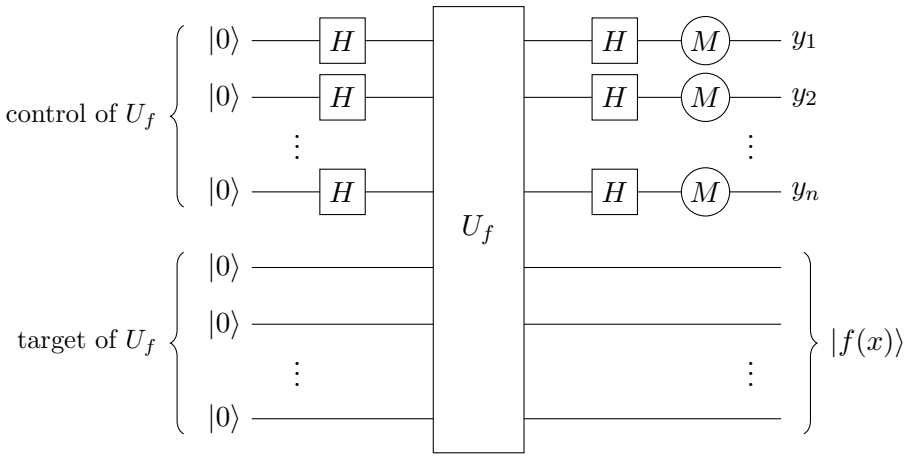


Figure 12: Simon's circuit. It makes one query and measures the n control qubits, giving a string y. The target register is not measured.

24.2 Analysis of One Run

For a state $\sum_{y,z} \alpha_{y,z}|y\rangle|z\rangle$ of the two registers, measuring only the first register gives y with probability $\sum_z |\alpha_{y,z}|^2$ (the probabilities of all the basis states $|y\rangle|z\rangle$ with this y add up).

Theorem 19 ▶ One Run of Simon's Circuit

Let y be the string measured by the circuit of Figure 12.

1. If f is 2-to-1 with period $s \neq 0^n$, then y is uniformly distributed over the 2^{n-1} strings y with $s \cdot y = 0$: each of them has probability $1/2^{n-1}$, and every y with $s \cdot y = 1$ has probability 0.
2. If f is 1-to-1, then y is uniformly distributed over all of $\{0,1\}^n$: each string has probability $1/2^n$.

Proof. We follow the state through the circuit. After the first layer of Hadamard gates, by Lemma 15,

$$\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle|0^n\rangle.$$

After the query U_f,

$$\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle|f(x)\rangle.$$

After the second layer of Hadamard gates on the control register, by Lemma 15 applied to every $|x\rangle$,

$$\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} \frac{1}{\sqrt{2^n}} \sum_{y\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle|f(x)\rangle = \sum_{y\in\{0,1\}^n} \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{x\cdot y}|y\rangle|f(x)\rangle. \tag{3}$$

The target register now carries $f(x)$, so to read off the amplitude of a basis state $|y\rangle|z\rangle$ we have to collect the x with $f(x) = z$.

Case 2: f is 1-to-1. Then the strings $f(x)$, $x \in \{0,1\}^n$, are all different, so every basis state $|y\rangle|f(x)\rangle$ in Equation (3) appears exactly once, with amplitude $\frac{1}{2^n}(-1)^{x\cdot y}$. Measuring the first register gives y with probability

$$\sum_{x\in\{0,1\}^n} \left|\frac{1}{2^n}(-1)^{x\cdot y}\right|^2 = 2^n \cdot \frac{1}{2^{2n}} = \frac{1}{2^n}.$$

Every string y is equally likely, and these 2^n probabilities add up to 1.

Case 1: f is 2-to-1 with period $s \neq 0^n$. Let R be the range of f; it has $|R| = 2^n/2 = 2^{n-1}$ elements. For every $z \in R$ pick one of its two preimages and call it x_z; the other preimage is $x'_z = x_z \oplus s$. Collecting in Equation (3) the two x with $f(x) = z$, the state is

$$\sum_{y\in\{0,1\}^n} \frac{1}{2^n} \sum_{z\in R} \Big((-1)^{x_z\cdot y} + (-1)^{(x_z\oplus s)\cdot y} \Big) |y\rangle|z\rangle = \sum_{y\in\{0,1\}^n} \frac{1}{2^n} \sum_{z\in R} (-1)^{x_z\cdot y} (1 + (-1)^{s\cdot y}) |y\rangle|z\rangle,$$

since $(x_z \oplus s)\cdot y = x_z \cdot y \oplus s \cdot y$. The amplitude of $|y\rangle|z\rangle$ is therefore $\frac{1}{2^n}(-1)^{x_z\cdot y}(1 + (-1)^{s\cdot y})$, whose absolute value does not depend on the sign $(-1)^{x_z\cdot y}$, and measuring the first register gives y with probability

$$\sum_{z\in R} \left| \frac{1}{2^n}(1 + (-1)^{s\cdot y}) \right|^2 = 2^{n-1} \cdot \frac{(1 + (-1)^{s\cdot y})^2}{2^{2n}} = \begin{cases} 2^{n-1} \cdot \dfrac{4}{2^{2n}} = \dfrac{2}{2^n} = \dfrac{1}{2^{n-1}} & \text{if } s\cdot y = 0, \\ 0 & \text{if } s\cdot y = 1. \end{cases}$$

The strings y with $s \cdot y = 0$ form the solution set of one nontrivial linear equation over $\{0,1\}$, so there are 2^{n-1} of them (the pairing $y \mapsto y \oplus e_i$ of the proof of Lemma 17, with $s_i = 1$, matches every y with $s \cdot y = 0$ to a y' with $s \cdot y' = 1$), and their probabilities add up to $2^{n-1} \cdot 1/2^{n-1} = 1$. □

Note that the second register was needed to *entangle* x with $f(x)$, so that only the two preimages of the same z interfere; we never look at it.

24.3 From the Measurements to s

Each run of the circuit gives one piece of information about s: a random y with $s \cdot y = 0$, i.e., one linear equation

$$s_1y_1 \oplus s_2y_2 \oplus \cdots \oplus s_ny_n = 0$$

for the unknown bits $s_1, \ldots, s_n$. (In class the count was "n measurements give n equations in n unknowns"; since s itself satisfies every equation, at most $n-1$ of them can be independent, and $n-1$ independent ones are enough, so we run the circuit $n-1$ times. One more run would not hurt.) Running the circuit $n-1$ times gives $n-1$ equations

$$y_1 \cdot s = 0, \quad y_2 \cdot s = 0, \quad \ldots, \quad y_{n-1} \cdot s = 0$$

in the n unknowns $s_1, \ldots, s_n$, a homogeneous system of linear equations over the field $\{0,1\}$ (addition is $\oplus$). If the $y_1, \ldots, y_{n-1}$ are *linearly independent* over $\{0,1\}$, the system has rank $n-1$ and its solution space has dimension $n - (n-1) = 1$, i.e., exactly two solutions: 0^n and one nonzero string s'. Both the test for linear independence and the solution s' are computed by Gaussian elimination over $\{0,1\}$ in $\mathcal{O}(n^3)$ time. Since $s \cdot y_i = 0$ for every i, the true s is always a solution; so if the y_i are independent and f is 2-to-1, then $s' = s$.

Algorithm 1 Simon's algorithm

```
run the circuit of Figure 12 n − 1 times; let y_1, ..., y_{n−1} be the measured strings
if y_1, ..., y_{n−1} are linearly dependent over {0,1}:    ▷ Gaussian elimination
    return FAIL
let s' be the unique nonzero solution of y_1 · s' = ··· = y_{n−1} · s' = 0
if f(s') = f(0^n):    ▷ two classical queries
    return s'    ▷ f is 2-to-1 with period s'
else:
    return 0^n    ▷ f is 1-to-1
```

The probability that the y_i are linearly independent will be shown in the next lecture (Theorem 29) to be, in the 2-to-1 case,

$$\left(1-\frac{1}{2}\right)\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{2^3}\right)\cdots\left(1-\frac{1}{2^{n-1}}\right) \geqslant \frac{1}{4},$$

and even larger (at least 1/2) in the 1-to-1 case. This gives:

Theorem 20 ▶ Simon's Algorithm

Algorithm 1 makes $n+1$ queries ($n-1$ quantum, 2 classical) and $\mathcal{O}(n^3)$ other operations. It never outputs a wrong answer, and it outputs the correct s with probability at least 1/4. Repeating it until it does not fail, t repetitions fail with probability at most $(3/4)^t$; so $\mathcal{O}(n)$ queries find s with constant probability and $\mathcal{O}(n \log n)$ queries find it with probability $1 - 1/n$.

Proof. Suppose the algorithm does not fail, so $y_1, \ldots, y_{n-1}$ are independent and s' is the unique nonzero solution of the system. If f is 2-to-1 with period s, then s is a nonzero solution, so $s' = s$, and the test $f(s') = f(0^n) = f(0^n \oplus s)$ succeeds: the output $s' = s$ is correct. If f is 1-to-1, then $f(s') \neq f(0^n)$ because $s' \neq 0^n$, and the output $0^n = s$ is correct. So the output is never wrong.

The algorithm fails only if the y_i are dependent. The $n-1$ runs of the circuit are independent (the register is re-initialized to $|0^n\rangle|0^n\rangle$ for every run). If f is 2-to-1, by Theorem 19 the y_i are independent uniform samples from the 2^{n-1} strings with $s \cdot y = 0$, and Theorem 29 shows that they are linearly independent with probability at least 1/4. If f is 1-to-1, the y_i are independent uniform samples from all of $\{0,1\}^n$, and Theorem 29 shows that they are linearly independent with probability at least 1/2. In both cases the success probability is at least 1/4, and t independent repetitions all fail with probability at most $(3/4)^t$. With $t = \left\lceil \log_{4/3} n \right\rceil$ repetitions this is at most $1/n$, and the number of queries is $t(n+1) = \mathcal{O}(n \log n)$. □

24.4 A Deterministic Classical Algorithm

The following was given as an exercise in class. It shows that, classically, $\mathcal{O}\left(\sqrt{2^n}\right)$ queries suffice even without randomness; the next lecture shows that $\Omega\left(\sqrt{2^n}\right)$ queries are also necessary, so the exercise is the best possible classical algorithm up to a constant factor.

Exercise 4. Find a deterministic classical algorithm for Simon's problem that makes $\mathcal{O}\left(\sqrt{2^n}\right)$ queries.

Solution. The idea is to query a set S of inputs such that every nonzero string s is the XOR of two elements of S. Then, if f is 2-to-1 with period s, those two elements collide ($f(x) = f(x \oplus s)$) and the collision reveals s.

Let $m = \lceil n/2 \rceil$ and split every n-bit string x as $x = (u, v)$ with $u \in \{0,1\}^m$ its first m bits and $v \in \{0,1\}^{n-m}$ its last $n-m$ bits. Let

$$S_1 = \{(u, 0^{n-m}) \mid u \in \{0,1\}^m\}, \qquad S_2 = \left\{(0^m, v) \mid v \in \{0,1\}^{n-m}\right\}, \qquad S = S_1 \cup S_2.$$

Then, since $n - m \leqslant m \leqslant (n+1)/2$, $|S| \leqslant 2^m + 2^{n-m} \leqslant 2 \cdot 2^m \leqslant 2 \cdot 2^{(n+1)/2} = 2\sqrt{2} \cdot \sqrt{2^n}$.

Claim: every $s \neq 0^n$ is the XOR of two different elements of S. Write $s = (u, v)$. Then $(u, 0^{n-m}) \in S_1$, $(0^m, v) \in S_2$ and $(u, 0^{n-m}) \oplus (0^m, v) = (u, v) = s$. The two elements are different, since otherwise $u = 0^m$ and $v = 0^{n-m}$, i.e., $s = 0^n$.

The algorithm. Query f on every $x \in S$ ($|S|$ queries). If there are $x \neq x'$ in S with $f(x) = f(x')$, output $s = x \oplus x'$; otherwise output $s = 0^n$.

Correctness. If f is 2-to-1 with period $s \neq 0^n$, by the claim there are $x \neq x'$ in S with $x \oplus x' = s$, i.e., $x' = x \oplus s$, so $f(x) = f(x')$ and the algorithm finds a collision. Any collision $f(x) = f(x')$ with $x \neq x'$ satisfies $x' = x \oplus s$ by the promise, so the output $x \oplus x' = s$ is correct whichever collision is found. If f is 1-to-1 there is no collision, and the output $0^n = s$ is correct.

The algorithm is deterministic, always correct, and makes $|S| \leqslant 2\sqrt{2} \cdot \sqrt{2^n} = \mathcal{O}\left(\sqrt{2^n}\right)$ queries. □

Lecture Date: October 1, 2026

Lecture Topic: Classical Query Complexity of Simon's Problem, Yao's Minmax Principle, Linear Independence in Simon's Algorithm

In this lecture, we determine the *classical* query complexity of Simon's problem. A randomized algorithm that queries random inputs and waits for a collision uses $\mathcal{O}(\sqrt{N \log N})$ queries, where $N = 2^n$, and we show with Yao's minmax principle that every randomized algorithm needs $\Omega\left(\sqrt{N}\right)$ queries. We then go through Simon's quantum algorithm once more, compute the measurement probabilities in both cases, and prove that the measured strings are linearly independent with probability at least $1/4$, which completes the analysis of the last lecture.

25 Simon's Problem, Classically

Recall Simon's problem: $f\colon \{0,1\}^n \to \{0,1\}^n$ is given as a black box, and we are promised that f is either 1-to-1, or 2-to-1 with a period $s \neq 0^n$, i.e.,

$$f(x) = f(y) \iff x = y \quad \text{or} \quad x \oplus y = s \text{ (i.e., } y = x \oplus s).$$

We have to find s (with $s = 0^n$ meaning that f is 1-to-1). Let $N = 2^n$. A classical algorithm learns f only through queries $f(q_1), f(q_2), \ldots$, and we may assume that it never queries the same input twice, since a repeated query gives no new information. The algorithm can find s only by finding a *collision*, two queries $q_i \neq q_j$ with $f(q_i) = f(q_j)$: by the promise this gives $s = q_i \oplus q_j$ immediately. Conversely, as long as all answers are distinct, the algorithm has no direct evidence of s.

25.1 Deterministic Algorithms

The exercise of the last lecture (Exercise 4) gives a deterministic algorithm with $\mathcal{O}\left(\sqrt{N}\right)$ queries. In class it was stated that this is optimal: every deterministic algorithm needs $\Omega\left(\sqrt{2^n}\right)$ queries. This is a special case of the lower bound for randomized algorithms proved below (Corollary 26).

25.2 A Randomized Algorithm

The natural randomized algorithm is the birthday paradox of the lecture of September 3: query random inputs and wait for a collision.

Algorithm 2 Randomized classical algorithm for Simon's problem

choose k distinct inputs $q_1, \ldots, q_k$ uniformly at random (without replacement) and query them
if $f(q_i) = f(q_j)$ for some $i \neq j$:
    return $q_i \oplus q_j$
else:
    return 0^n

Theorem 21 ▶ Randomized Upper Bound

If f is 1-to-1, Algorithm 2 is always correct. If f is 2-to-1 with period s, it is wrong with probability at most $e^{-k(k-1)/(2N)}$. In particular, with $k \geqslant \sqrt{2N \ln N} + 1$ queries (an integer at most N once $n \geqslant 3$) the algorithm is correct with probability at least $1 - 1/N$, and with $k \geqslant \sqrt{2N} + 1$, i.e., $\mathcal{O}\left(\sqrt{N}\right)$, queries with probability at least $1 - 1/e$.

Proof. If f is 1-to-1 there is no collision, and the output 0^n is correct. Let f be 2-to-1 with period $s \neq 0^n$. Any collision $f(q_i) = f(q_j)$ gives $q_i \oplus q_j = s$ by the promise, so the algorithm is wrong only if no collision occurs among the k queries. Call $q_j \oplus s$ the *mirror* of q_j; a collision occurs at step i exactly when q_i is the mirror of an earlier query q_j, $j < i$.

Suppose that no collision occurred among $q_1, \ldots, q_{i-1}$. The $i-1$ mirrors $q_1 \oplus s, \ldots, q_{i-1} \oplus s$ are distinct (because the q_j are distinct), and none of them is among $q_1, \ldots, q_{i-1}$ (that would be a collision). So all $i-1$ mirrors are among the $N - (i-1)$ inputs that have not been queried, and q_i is uniform among those. Hence

$$\mathbb{P}(\text{collision at step } i \mid \text{no collision before}) = \frac{i-1}{N-i+1} \geqslant \frac{i-1}{N},$$

and the probability of no collision at step i, given none before, is at most $1 - (i-1)/N$. These are the factors $1, 1 - 1/N, 1 - 2/N, \ldots$ in the notes. Multiplying over the k steps and using $1 - x \leqslant e^{-x}$,

$$\begin{aligned}\mathbb{P}(\text{no collision among } q_1, \ldots, q_k) &\leqslant \prod_{i=1}^{k} \left(1 - \frac{i-1}{N}\right) \leqslant \prod_{i=1}^{k} e^{-(i-1)/N} \\ &= e^{-\sum_{i=1}^{k}(i-1)/N} = e^{-k(k-1)/(2N)}.\end{aligned}$$

With $k - 1 \geqslant \sqrt{2N \ln N}$ this is at most $e^{-(k-1)^2/(2N)} \leqslant e^{-\ln N} = 1/N$, and with $k - 1 \geqslant \sqrt{2N}$ it is at most e^{-1}. □

So $\mathcal{O}\left(\sqrt{N}\right)$ queries give a constant success probability and $\mathcal{O}(\sqrt{N \log N})$ queries give success probability $1 - 1/N$ ("with high probability"). The next theorem shows that randomness cannot do much better than this.

25.3 Lower Bound for Randomized Algorithms via Yao's Principle

Theorem 22 ▶ Randomized Lower Bound

Let $n \geqslant 2$. Every randomized classical algorithm that solves Simon's problem with success probability at least $2/3$ on every input makes more than $\sqrt{N}/2$ queries on some input. Hence every such algorithm needs $\Omega\left(\sqrt{N}\right) = \Omega\left(\sqrt{2^n}\right)$ queries.

The proof uses Yao's minmax principle from the lecture of September 1: to lower bound randomized algorithms on their worst input, it is enough to pick a *distribution* μ on the inputs and to lower bound the error of every *deterministic* algorithm on an input drawn from μ. On September 1 the principle was stated for the expected number of queries of Las Vegas algorithms; here we need the version for the error probability of algorithms with a fixed number of queries, and only its easy direction, which is an averaging argument:

Lemma 23 ▶ Yao's Minmax Principle, Easy Direction

Let μ be a probability distribution on the inputs. If every deterministic algorithm that makes at most k queries errs with probability greater than δ on an input drawn from μ, then every randomized algorithm that makes at most k queries errs with probability greater than δ on some input.

Proof. A randomized algorithm R is a deterministic algorithm A_r together with a random string r: once r is fixed, A_r is deterministic and makes at most k queries. Let $\mathrm{err}(A, x) \in \{0, 1\}$ indicate that A is wrong on input x. Then

$$\max_x \mathbb{P}_r(\mathrm{err}(A_r, x) = 1) \geqslant \mathbb{E}_{x\sim\mu}[\mathbb{P}_r(\mathrm{err}(A_r, x) = 1)] = \mathbb{E}_r[\mathbb{P}_{x\sim\mu}(\mathrm{err}(A_r, x) = 1)] > \delta,$$

since a maximum is at least an average, the order of the two expectations can be exchanged, and every inner probability on the right is greater than δ by assumption. So on the input x achieving the maximum, R errs with probability greater than δ. □

The input distribution. We use the distribution μ of Figure 13:

- with probability $1/2$, f is a uniformly random 1-to-1 function, i.e., the table of values $(f(0^n), \ldots, f(1^n))$ is a uniformly random permutation of $\{0, 1\}^n$ (call this distribution $\mathcal{P}$);
- with probability $1/2$, f is drawn from the distribution $\mathcal{D}$ of random 2-to-1 functions: choose s uniformly among the $N - 1$ nonzero strings, which splits $\{0, 1\}^n$ into $N/2$ pairs $\{x, x \oplus s\}$; then assign to the pairs distinct values, uniformly at random among all such assignments, and let $f(x) = f(x \oplus s)$ be the value of the pair of x.

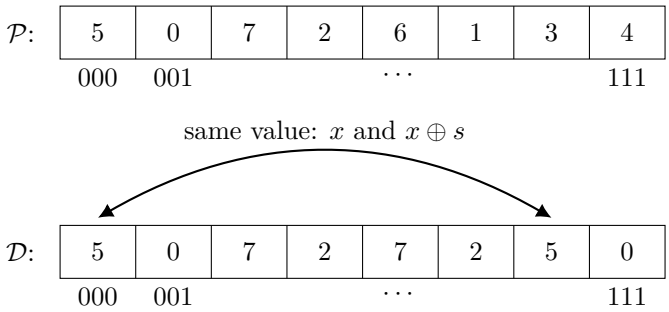


Figure 13: The two halves of the input distribution μ for $n = 3$ (values written as integers). Under $\mathcal{P}$ the table is a random permutation. Under $\mathcal{D}$ a random nonzero s (here $s = 110$) pairs up the inputs, and the two inputs of a pair get the same random value, different pairs different values.

Principle of deferred decisions. We do not have to choose the whole table in advance. We can generate the random input lazily, while the algorithm runs, and only decide the values that the algorithm asks for; the distribution of the answers is the same.

- Under $\mathcal{P}$: when the algorithm queries a new input q_i, the answer $f(q_i)$ is a uniformly random value among the $N - (i - 1)$ values not used by the earlier answers.
- Under $\mathcal{D}$: choose s uniformly among the nonzero strings and keep it hidden. When the algorithm queries q_i: if $q_i = q_j \oplus s$ for an earlier query q_j (a *collision*: q_i is the mirror of q_j), the answer is the earlier answer $f(q_j)$; otherwise q_i lies in a new pair, and the answer is a uniformly random value among the values not used so far ($N - (i - 1)$ of them as long as no collision has occurred).

Both procedures produce exactly the distributions $\mathcal{P}$ and $\mathcal{D}$ on the queried values, because a uniformly random injective assignment of values can be built one value at a time, each new value uniform among the unused ones.

The lower bound has two steps: (1) under $\mathcal{D}$, an algorithm with few queries is unlikely to see a collision; (2) as long as it sees no collision, its answers look exactly as under $\mathcal{P}$, so it cannot tell the two cases apart. Step (1) is the computation done in class:

Lemma 24 ▶ Few Queries Rarely Collide

Let $n \geqslant 2$, let $0 < \epsilon \leqslant 1/2$, and let A be a deterministic algorithm that makes at most $k \leqslant \epsilon\sqrt{N}$ queries. Under $\mathcal{D}$, the probability that two of the queries of A form a pair $\{x, x \oplus s\}$ (a collision) is at most

$$\sum_{i=1}^{k} \frac{i-1}{N-1-\binom{i-1}{2}} \leqslant \frac{k(k-1)}{N} < \epsilon^2.$$

Proof. Let Q_i be the event that there is no collision among the first i queries, and let C_i be the event that the ith query collides with an earlier one, i.e., $q_i \oplus s \in \{q_1, \ldots, q_{i-1}\}$. A collision among the k queries means that some C_i occurs, and if C_i is the first one then Q_{i-1} holds. So by the union bound

$$\mathbb{P}(\text{collision}) \leqslant \sum_{i=1}^{k} \mathbb{P}(C_i \cap Q_{i-1}).$$

Fix i and condition on Q_{i-1} and on the answers $f(q_1), \ldots, f(q_{i-1})$ seen so far (they are distinct, as Q_{i-1} holds). What does the algorithm know about s? Since A is deterministic, the queries $q_1, \ldots, q_{i-1}$ are the same for every s that leads to these answers. By the lazy generation above, the answers were $i-1$ fresh random values, and this particular sequence of answers has probability $\prod_{j=1}^{i-1} 1/(N-j+1)$, the same for every s that is not of the form $q_j \oplus q_l$ with $j < l \leqslant i-1$; for an s of this form, the queries q_j and q_l would have collided, contradicting Q_{i-1}, so the probability is 0. By Bayes' rule, given the answers, s is uniformly distributed over the nonzero strings that are not among the at most $\binom{i-1}{2}$ strings $q_j \oplus q_l$, a set of at least $N-1-\binom{i-1}{2}$ candidates. The next query q_i is determined by the answers (A is deterministic), and C_i occurs exactly when $s \in \{q_i \oplus q_1, \ldots, q_i \oplus q_{i-1}\}$, a set of $i-1$ strings. Therefore

$$\mathbb{P}(C_i \mid Q_{i-1}, \text{answers}) \leqslant \frac{i-1}{N-1-\binom{i-1}{2}},$$

and the same bound holds for $\mathbb{P}(C_i \cap Q_{i-1})$ after averaging over the answers.

For the final estimate, $\binom{i-1}{2} \leqslant \binom{k}{2} \leqslant k^2/2 \leqslant \epsilon^2 N/2 \leqslant N/8$, so the denominator is at least $N-1-N/8 \geqslant N/2$ for $N \geqslant 4$. Hence

$$\mathbb{P}(\text{collision}) \leqslant \sum_{i=1}^{k} \frac{2(i-1)}{N} = \frac{2}{N} \cdot \frac{k(k-1)}{2} = \frac{k(k-1)}{N} < \frac{k^2}{N} \leqslant \epsilon^2. \qquad \square$$

Step (2) says that, without a collision, the algorithm sees the same thing under both distributions.

Lemma 25 ▶ Indistinguishability

Let A, k and ϵ be as in Lemma 24. For every sequence t of answers (the *transcript* $T = (f(q_1), \ldots, f(q_k))$ of A),

$$\mathbb{P}_{\mathcal{D}}(T = t) \geqslant \mathbb{P}_{\mathcal{D}}(T = t \text{ and no collision}) \geqslant (1-\epsilon^2)\,\mathbb{P}_{\mathcal{P}}(T = t).$$

Proof. If t has two equal entries then $\mathbb{P}_{\mathcal{P}}(T = t) = 0$ and there is nothing to show. Let $t = (a_1, \ldots, a_k)$ have distinct entries. Under $\mathcal{P}$ every answer is uniform among the unused values, so

$$\mathbb{P}_{\mathcal{P}}(T = t) = \prod_{i=1}^{k} \frac{1}{N - i + 1}.$$

Under $\mathcal{D}$, write p_i for the probability of a collision at step i given Q_{i-1} and the answers $a_1, \ldots, a_{i-1}$; by the proof of Lemma 24, $p_i \leqslant \pi_i := (i-1)/(N - 1 - \binom{i-1}{2})$. If there is no collision at step i, the answer is a fresh value, uniform among the $N - i + 1$ unused ones, so it equals a_i with probability $1/(N - i + 1)$. Multiplying over the steps,

$$\begin{aligned}\mathbb{P}_{\mathcal{D}}(T = t \text{ and no collision}) &= \prod_{i=1}^{k}(1 - p_i)\frac{1}{N - i + 1} \geqslant \prod_{i=1}^{k}(1 - \pi_i) \cdot \mathbb{P}_{\mathcal{P}}(T = t) \\ &\geqslant \left(1 - \sum_{i=1}^{k} \pi_i\right) \mathbb{P}_{\mathcal{P}}(T = t),\end{aligned}$$

using $\prod(1 - \pi_i) \geqslant 1 - \sum \pi_i$ for $\pi_i \in [0, 1]$ (Lemma 28 below; $\pi_i \leqslant 2(i-1)/N \leqslant 1$ by the estimate in the proof of Lemma 24). By Lemma 24, $\sum_i \pi_i < \epsilon^2$. □

Proof of Theorem 22. Let A be a deterministic algorithm with at most $k \leqslant \sqrt{N}/2$ queries, i.e., $\epsilon = 1/2$ in the lemmas. We may assume that A makes exactly k queries (extra queries whose answers are ignored change nothing), so its transcript T has length k. The output of A is a function of T. Let $\mathcal{T}$ be the set of transcripts on which A outputs 0^n ("f is 1-to-1"), and let $p = \mathbb{P}_{\mathcal{P}}(T \in \mathcal{T})$ be the probability that A is correct on a random 1-to-1 function. Under $\mathcal{D}$, A can only be correct if it does not output 0^n, so by Lemma 25 summed over all $t \in \mathcal{T}$,

$$\mathbb{P}_{\mathcal{D}}(A \text{ correct}) \leqslant \mathbb{P}_{\mathcal{D}}(T \notin \mathcal{T}) = 1 - \mathbb{P}_{\mathcal{D}}(T \in \mathcal{T}) \leqslant 1 - (1 - \epsilon^2)\,\mathbb{P}_{\mathcal{P}}(T \in \mathcal{T}) = 1 - (1 - \epsilon^2)p.$$

Therefore the success probability of A on an input drawn from μ is

$$\frac{1}{2}p + \frac{1}{2}\Big(1 - (1 - \epsilon^2)p\Big) = \frac{1}{2} + \frac{\epsilon^2}{2}p \leqslant \frac{1}{2} + \frac{\epsilon^2}{2} = \frac{1}{2} + \frac{1}{8} = \frac{5}{8} < \frac{2}{3}.$$

So every deterministic algorithm with at most $\sqrt{N}/2$ queries errs with probability at least $3/8 > 1/3$ on μ. By Lemma 23 (with $\delta = 1/3$), every randomized algorithm with at most $\sqrt{N}/2$ queries errs with probability greater than $1/3$ on some input, i.e., its success probability on that input is less than $2/3$. Hence an algorithm with success probability at least $2/3$ on every input must make more than $\sqrt{N}/2$ queries on some input. □

In words, as it was put in class: on the first distribution all answers are distinct and the algorithm cannot solve the problem quickly; on the second distribution the answers are also all distinct unless a collision occurs, which is unlikely with few queries, so the algorithm cannot quickly find any difference between the two distributions. The success probability $2/3$ is not important: the same proof with $\epsilon = \sqrt{2\gamma}$ shows that, for $\gamma \leqslant 1/8$, success probability $1/2 + \gamma$ needs more than $\sqrt{2\gamma N}$ queries.

Corollary 26 ▶ Deterministic Lower Bound

Every deterministic classical algorithm for Simon's problem makes more than $\sqrt{N}/2$ queries on some input. Hence the deterministic query complexity of Simon's problem is $\Theta\left(\sqrt{N}\right)$.

Proof. A deterministic algorithm that is always correct is a randomized algorithm (one that ignores its random bits) with success probability $1 \geqslant 2/3$ on every input, so Theorem 22 applies to it. The matching upper bound is Exercise 4. □

Corollary 27 ▶ Classical Query Complexity of Simon's Problem

The classical query complexity of Simon's problem is $\Theta\left(\sqrt{N}\right) = \Theta\left(\sqrt{2^n}\right)$, both for deterministic algorithms (Exercise 4, Corollary 26) and for randomized algorithms with constant success probability (Theorem 21, Theorem 22). Simon's quantum algorithm, with $\mathcal{O}(n)$ queries, is therefore exponentially faster than every classical algorithm.

We used the following elementary inequality.

Lemma 28

For $a_1, \ldots, a_m \in [0,1]$, $\prod_{i=1}^{m}(1-a_i) \geqslant 1 - \sum_{i=1}^{m} a_i$.

Proof. By induction on m; for $m = 1$ it is an equality. If it holds for $m-1$, then, multiplying the induction hypothesis by $1 - a_m \geqslant 0$ and using $a_m \sum_{i<m} a_i \geqslant 0$ in the last step,

$$\prod_{i=1}^{m}(1-a_i) \geqslant \left(1 - \sum_{i=1}^{m-1} a_i\right)(1-a_m) = 1 - \sum_{i=1}^{m-1} a_i - a_m + a_m \sum_{i=1}^{m-1} a_i \geqslant 1 - \sum_{i=1}^{m} a_i. \qquad \square$$

26 Simon's Algorithm Revisited

In class the quantum algorithm was gone through again (Figure 12 of the last lecture) with the measurement probabilities written out; the computations are those of Theorem 19, so we only record what is new. Just before the measurement the state is

$$\sum_{y\in\{0,1\}^n} \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{x\cdot y} |f(x)\rangle|y\rangle$$

(target register written first), so for a fixed y the probability of measuring y in the control register is the squared norm of the vector that multiplies $|y\rangle$,

$$\mathbb{P}(y) = \left\| \frac{1}{2^n} \sum_{x\in\{0,1\}^n} (-1)^{x\cdot y} |f(x)\rangle \right\|_2^2,$$

because the probability of a measurement outcome is the sum of the squared amplitudes of all basis states $|z\rangle|y\rangle$ with this y.

26.1 The 1-to-1 Case

In class the computation for a 1-to-1 function ended with a question mark: the "measure" (the probability of y) was computed as $\frac{1}{2^{2n}} \cdot 2^n = \frac{1}{2^n}$, and it was asked whether this is right and whether the total is 1.

Exercise 5. Suppose that f is 1-to-1. What is the probability of measuring a given string y in Simon's circuit, and do these probabilities add up to 1?

Solution. If f is 1-to-1, the strings $f(x)$ for the 2^n different x are 2^n different basis states, so we may replace $|f(x)\rangle$ by $|x\rangle$ (this only renames the basis states and does not change the norm). The vector multiplying $|y\rangle$ is then $\frac{1}{2^n}\sum_x (-1)^{x\cdot y}|x\rangle$, which has 2^n orthonormal components, each of absolute value $1/2^n$. Its squared norm is

$$\mathbb{P}(y) = \sum_{x\in\{0,1\}^n} \left(\frac{1}{2^n}\right)^2 = 2^n \cdot \frac{1}{2^{2n}} = \frac{1}{2^n}.$$

This is the squared norm, not the norm (the norm would be $1/\sqrt{2^n}$); the probability of an outcome is always the squared norm. Every one of the 2^n strings y has the same probability $1/2^n$, so the measured y is uniformly distributed over $\{0,1\}^n$, and the total probability is $2^n \cdot 1/2^n = 1$, as it must be. □

26.2 The 2-to-1 Case

For f 2-to-1 with period $s \neq 0^n$, collecting the two preimages x_z and $x_z \oplus s$ of every z in the range R of f turns the vector into $\frac{1}{2^n}\sum_{z\in R}(-1)^{x_z\cdot y}(1+(-1)^{s\cdot y})|z\rangle$, whose squared norm is $2^{n-1}\cdot 4/2^{2n} = 1/2^{n-1}$ if $s\cdot y = 0$ and 0 if $s\cdot y = 1$ (Theorem 19, case 1). So we only ever measure strings y with $s\cdot y = 0$, each of the 2^{n-1} such strings equally likely, and the total is $2^{n-1}\cdot 1/2^{n-1} = 1$.

26.3 Solving for s

Running the circuit $n-1$ times gives strings $y_1,\dots,y_{n-1}$, and in the 2-to-1 case the system of linear equations over $\{0,1\}$

$$s\cdot y_1 = 0, \quad s\cdot y_2 = 0, \quad \dots, \quad s\cdot y_{n-1} = 0.$$

We want the y_i to be *linearly independent*: then the $(n-1)\times n$ coefficient matrix has rank $n-1$, and Gaussian elimination gives a one-dimensional solution space $\{0^n, s'\}$ with a unique nonzero solution s'. For the nonzero solution s' of the system we test $f(0^n) = f(s')$ with two classical queries: if the test is true, then f is 2-to-1 and $s = s'$ (by the promise, $f(0^n) = f(s')$ with $s' \neq 0^n$ means $s' = 0^n \oplus s$); if it is false, then f is 1-to-1 (in the 2-to-1 case s itself solves the system, so $s' = s$ and the test would be true). This is Algorithm 1 of the last lecture. It remains to show the claim about linear independence that was left open there.

Exercise 6. Show that the probability that the y's are linearly independent is at least $1/4$.

In the 2-to-1 case the y_i are independent uniform samples from $V = \{y \mid s\cdot y = 0\}$, which is a subspace of $\{0,1\}^n$ (over the field with two elements) of dimension $n-1$, with 2^{n-1} elements. The following theorem answers the exercise, and also covers the 1-to-1 case, where the y_i are uniform samples from the whole space $\{0,1\}^n$ of dimension n.

Theorem 29 ▶ Linear Independence of Random Vectors

Let V be a subspace of $\{0,1\}^n$ of dimension $d \geqslant 1$, so $|V| = 2^d$, and let $y_1, y_2, \dots$ be independent uniformly random elements of V.

1. $y_1,\dots,y_d$ are linearly independent with probability

$$\prod_{i=1}^{d}\left(1-\frac{1}{2^i}\right) = \left(1-\frac{1}{2}\right)\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{2^3}\right)\cdots\left(1-\frac{1}{2^d}\right) \geqslant \frac{1}{4}.$$

2. $y_1,\dots,y_{d-1}$ are linearly independent with probability $\prod_{i=2}^{d}(1-1/2^i) \geqslant 1/2$.

In particular, in the 2-to-1 case ($V = \{y \mid s\cdot y = 0\}$, $d = n-1$) the strings $y_1,\dots,y_{n-1}$ of Simon's algorithm are linearly independent with probability at least $1/4$, and in the 1-to-1 case ($V = \{0,1\}^n$, $d = n$) with probability at least $1/2$.

Proof. Suppose that $y_1,\dots,y_j$ are linearly independent, $0 \leqslant j < d$. Their span consists of the 2^j vectors $\sum_{i\leqslant j} c_i y_i$ with $c_i \in \{0,1\}$, all different (a nontrivial combination that gives 0 would be a

linear dependence), and it lies inside V because V is a subspace. The vector y_{j+1} is uniform over the 2^d elements of V, independent of $y_1, \ldots, y_j$, and $y_1, \ldots, y_{j+1}$ are linearly independent exactly when y_{j+1} is not in this span. So, given that $y_1, \ldots, y_j$ are independent, the probability that $y_1, \ldots, y_{j+1}$ are independent is

$$1 - \frac{2^j}{2^d} = 1 - \frac{1}{2^{d-j}}.$$

(This is the factor "$1 - 2^k/2^n$" written in class, with the 2^n replaced by the size 2^d of the space the y's live in.) Multiplying these conditional probabilities for $j = 0, 1, \ldots, m-1$, the first m vectors are linearly independent with probability

$$\prod_{j=0}^{m-1} \left(1 - \frac{1}{2^{d-j}}\right) = \prod_{i=d-m+1}^{d} \left(1 - \frac{1}{2^i}\right).$$

For $m = d$ this is $\prod_{i=1}^{d}(1 - 2^{-i})$ and for $m = d - 1$ it is $\prod_{i=2}^{d}(1 - 2^{-i})$.

For the lower bounds we use Lemma 28 and the geometric series $\sum_{i \geqslant 2} 2^{-i} = 1/2$:

$$\prod_{i=2}^{d} \left(1 - \frac{1}{2^i}\right) \geqslant 1 - \sum_{i=2}^{d} \frac{1}{2^i} \geqslant 1 - \frac{1}{2} = \frac{1}{2},$$

which proves the second claim, and then

$$\prod_{i=1}^{d} \left(1 - \frac{1}{2^i}\right) = \frac{1}{2} \prod_{i=2}^{d} \left(1 - \frac{1}{2^i}\right) \geqslant \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

which proves the first. The claims about Simon's algorithm follow from the measurement distributions computed above: in the 2-to-1 case each y_i is uniform over the 2^{n-1} strings with $s \cdot y = 0$, which form a subspace of dimension $d = n - 1$ (the solution set of one nontrivial homogeneous linear equation), and we take $n - 1 = d$ samples; in the 1-to-1 case each y_i is uniform over $\{0,1\}^n$, $d = n$, and we take $n - 1 = d - 1$ samples. □

Note that we need the first factor $1/2$ to be pulled out before using Lemma 28: the bound $1 - \sum_{i \geqslant 1} 2^{-i} = 0$ would be useless. The infinite product $\prod_{i \geqslant 1}(1 - 2^{-i}) = 0.2887\ldots$, so the constant $1/4$ cannot be improved much, and it does not depend on n. Together with Theorem 20 this completes the analysis of Simon's algorithm: $n - 1$ runs of the circuit plus two classical queries find s with probability at least $1/4$, a constant number of repetitions makes the success probability any constant we like, and $\mathcal{O}(\log n)$ repetitions make it $1 - 1/n$.

Lecture Date: October 6, 2026

Lecture Topic: Grover's Search Algorithm, Phase Flip, Diffusion Operator, Rotation Analysis

In this lecture, we study *Grover's search algorithm*, which finds a solution of an unstructured search problem over $N = 2^n$ inputs with $\mathcal{O}\left(\sqrt{N}\right)$ quantum queries, whereas every classical algorithm needs $\Omega(N)$ queries. We first run the algorithm by hand on two qubits, then write it for n qubits using two operators, the *phase flip* Z_f and the *diffusion operator* D, and finally analyze it exactly: every Grover iteration is a rotation by a fixed angle 2θ in a two-dimensional plane, which tells us precisely how many iterations to make.

27 Unstructured Search

Problem 5 ▶ Unstructured Search

Given $f\colon \{0,1\}^n \to \{0,1\}$ as a black box, find an x with $f(x) = 1$.

Think of f as a Boolean formula and of x as an assignment: $f(x) = 1$ means that x satisfies the formula. In *unique satisfiability* we are promised that there is exactly one x with $f(x) = 1$. More generally, let

$$a = |\{x \mid f(x) = 1\}|, \qquad b = |\{x \mid f(x) \neq 1\}| = N - a$$

be the numbers of solutions and of non-solutions, where $N = 2^n$. The search is *unstructured*: we know nothing about f except what the queries tell us, so classically the best we can do is to try inputs one by one. For $a = 1$ every classical algorithm, deterministic or randomized, needs $\Omega(N)$ queries (lectures of September 1 and 3: random sampling needs N expected queries, and $\Omega(N)$ queries are necessary). Grover's algorithm needs only $\mathcal{O}\left(\sqrt{N}\right)$ queries, and more generally $\mathcal{O}\left(\sqrt{N/a}\right)$. (It is known, but not shown in this course, that $\Omega\left(\sqrt{N}\right)$ queries are necessary for every quantum algorithm, so Grover's algorithm is optimal.)

28 Grover's Algorithm on Two Qubits

Let $n = 2$, $N = 4$, and suppose that the unique solution is $x^* = 10$. We write a state of the two qubits as its vector of four amplitudes $(a_{00}, a_{01}, a_{10}, a_{11})$. The algorithm has four steps (Figure 14).

1. *Put the register in the uniform superposition with Hadamard gates:*

$$H^{\otimes 2}|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \left(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}\right).$$

2. *Phase flip.* Recreate f as a quantum circuit, the query $U_f\colon |x\rangle|c\rangle \mapsto |x\rangle|c \oplus f(x)\rangle$. With the target qubit in $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ this is the phase flip of the lecture of September 24,

$$U_f\colon |x\rangle \longmapsto (-1)^{f(x)}|x\rangle,$$

which flips the sign of the amplitude of every solution. One query gives

$$\left(\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}\right).$$

3. *Grover diffusion operator.* Let $\mu = \frac{1}{N}\sum_x a_x$ be the mean of the amplitudes. The diffusion operator D reflects every amplitude about the mean (*inversion about the mean*):

$$D\colon a_x \longmapsto 2\mu - a_x \qquad \text{for every } x.$$

Here $\mu = \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \frac{1}{2}\right) = \frac{1}{4}$, so the three non-solutions become $2 \cdot \frac{1}{4} - \frac{1}{2} = 0$ and the solution becomes $2 \cdot \frac{1}{4} + \frac{1}{2} = 1$:

$$\left(\tfrac{1}{2}, \tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{1}{2}\right) \longmapsto (0, 0, 1, 0) = |10\rangle.$$

Measuring now gives the solution $x^* = 10$ with probability 1, after a single query.

4. In general, *repeat steps 2 and 3* k times, where k is about $\sqrt{N}$ (we compute the exact k in Section 30); for $N = 4$ one iteration is enough.

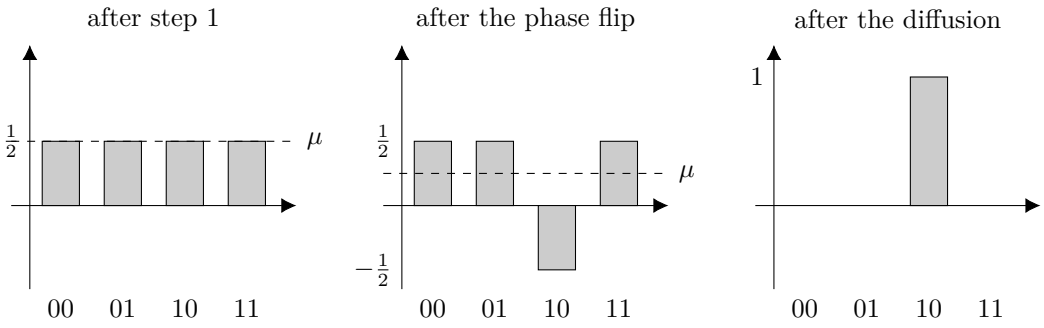


Figure 14: Grover's algorithm for $N = 4$ with the solution $x^* = 10$: the amplitudes after the Hadamard gates, after the phase flip (the mean drops to 1/4), and after the inversion about the mean. One iteration moves all the amplitude onto the solution.

The idea behind steps 2 and 3: the phase flip makes the solution's amplitude negative, which lowers the mean a little; the inversion about the mean then pushes the solution's amplitude up above the mean by the amount it was below, while all the other amplitudes shrink a little. Each iteration transfers amplitude from the non-solutions to the solution.

29 Grover's Algorithm on n Qubits

For n qubits and a unique solution x^* the same steps apply to the vector $(a_x)_{x\in\{0,1\}^n}$ of the N amplitudes. After the Hadamard gates every amplitude is $1/\sqrt{N}$, and the phase flip makes the amplitude of x^* equal to $-1/\sqrt{N}$. The mean is then

$$\mu = \frac{1}{N}\left(\frac{N-1}{\sqrt{N}} - \frac{1}{\sqrt{N}}\right) = \frac{N-2}{N\sqrt{N}} \approx \frac{1}{\sqrt{N}},$$

so the diffusion operator maps the amplitude of x^* to $2\mu + 1/\sqrt{N} \approx 3/\sqrt{N}$ and every other amplitude to $2\mu - 1/\sqrt{N} \approx 1/\sqrt{N}$. The next iteration takes x^* to $-3/\sqrt{N}$ and then to about $5/\sqrt{N}$, the one after to about $7/\sqrt{N}$, and so on: as long as the solution's amplitude is small, the mean stays close to $1/\sqrt{N}$ and each iteration adds about $2/\sqrt{N}$ to it, so after about $\sqrt{N}$ iterations the amplitude of x^* is a constant and a measurement finds x^* with constant probability. The exact analysis in Section 30 confirms this and gives the right number of iterations.

29.1 The Operators Z_f, Z_0 and D

We now write the two steps of a Grover iteration as unitary operators on n qubits.

The phase flip Z_f. The phase flip

$$Z_f\colon |x\rangle \longmapsto (-1)^{f(x)}|x\rangle$$

is implemented by the query U_f with an extra qubit in the state $|1\rangle$ passed through a Hadamard gate (Figure 15): the extra qubit is in $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ when U_f is applied, so U_f multiplies $|x\rangle$ by $(-1)^{f(x)}$ and leaves the extra qubit unchanged, and the second Hadamard gate returns the extra qubit to $|1\rangle$. One application of Z_f is one query.

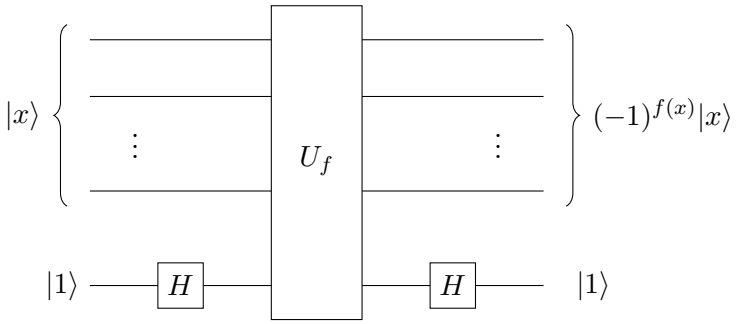


Figure 15: The phase flip Z_f: one query U_f with the extra qubit in $H|1\rangle$ flips the sign of every solution $|x\rangle$. The extra qubit comes back to $|1\rangle$ and can be reused.

The phase flip Z_0 about 0^n. Let $f_0\colon \{0,1\}^n \to \{0,1\}$ be the function with $f_0(x) = 1$ exactly for $x = 0^n$, i.e., $f_0(x) = \bar{x}_1 \wedge \bar{x}_2 \wedge \cdots \wedge \bar{x}_n$ where $\bar{x}_i = \neg x_i$. Its phase flip is

$$Z_0\colon |x\rangle \longmapsto \begin{cases} -|x\rangle & \text{if } x = 0^n, \\ |x\rangle & \text{otherwise,} \end{cases} \qquad \text{i.e.,} \qquad Z_0 = I - 2|0^n\rangle\langle 0^n|.$$

As a matrix, $|0^n\rangle\langle 0^n|$ is the $N \times N$ matrix with a single 1 in the top left corner, so

$$Z_0 = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & \\ \vdots & & \ddots & \\ 0 & & & 1 \end{pmatrix} - 2\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & 1 & & \\ \vdots & & \ddots & \\ 0 & & & 1 \end{pmatrix}.$$

Z_0 is implemented exactly like Z_f, with the circuit for f_0 in place of the black box: the reversible gate $|x\rangle|c\rangle \mapsto |x\rangle|c \oplus (\bar{x}_1 \wedge \cdots \wedge \bar{x}_n)\rangle$ (a Toffoli gate with n negated controls) and an extra qubit in $H|1\rangle$. It makes no query to f.

The diffusion operator D. Let $|h\rangle = H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle$ be the uniform superposition. The diffusion operator is

$$D = -H^{\otimes n} Z_0 H^{\otimes n}.$$

Lemma 30 ▶ Inversion About the Mean

$D = 2|h\rangle\langle h| - I$, and for every state $\sum_x a_x|x\rangle$ with mean amplitude $\mu = \frac{1}{N}\sum_x a_x$,

$$D \sum_{x\in\{0,1\}^n} a_x|x\rangle = \sum_{x\in\{0,1\}^n} (2\mu - a_x)|x\rangle.$$

Proof. Since $H^\dagger = H$ and $H^2 = I$, we have $H^{\otimes n}H^{\otimes n} = I$, and $(H^{\otimes n}|0^n\rangle)^\dagger = \langle 0^n|H^{\otimes n}$, i.e., $\langle 0^n|H^{\otimes n} = \langle h|$. Therefore

$$\begin{aligned} D &= -H^{\otimes n}(I - 2|0^n\rangle\langle 0^n|)H^{\otimes n} = -\big(H^{\otimes n}IH^{\otimes n} - 2H^{\otimes n}|0^n\rangle\langle 0^n|H^{\otimes n}\big) \\ &= -(I - 2|h\rangle\langle h|) = 2|h\rangle\langle h| - I. \end{aligned}$$

For the second claim, $\langle h|\psi\rangle = \frac{1}{\sqrt{N}}\sum_x a_x = \sqrt{N}\mu$ for $|\psi\rangle = \sum_x a_x|x\rangle$, so

$$2|h\rangle\langle h|\psi\rangle = 2\sqrt{N}\mu|h\rangle = 2\mu\sum_x|x\rangle, \qquad \text{and} \qquad D|\psi\rangle = 2\mu\sum_x|x\rangle - \sum_x a_x|x\rangle = \sum_x(2\mu - a_x)|x\rangle.$$

□

So D is exactly the inversion about the mean of step 3, and it costs no query. (The notes write $D = I - 2|h\rangle\langle h|$, which is $-D$; the overall sign of D only multiplies the state after k iterations by $(-1)^k$, a global phase that changes no measurement probability. We keep the sign that makes D the inversion about the mean.)

The Grover iteration. One iteration of steps 2 and 3 is the operator

$$G = DZ_f = -H^{\otimes n}Z_0H^{\otimes n}Z_f,$$

and Grover's algorithm is: prepare $|h\rangle = H^{\otimes n}|0^n\rangle$, apply G k times, measure (Figure 16). It makes k queries, one per iteration; the Hadamard gates and Z_0 make no query.

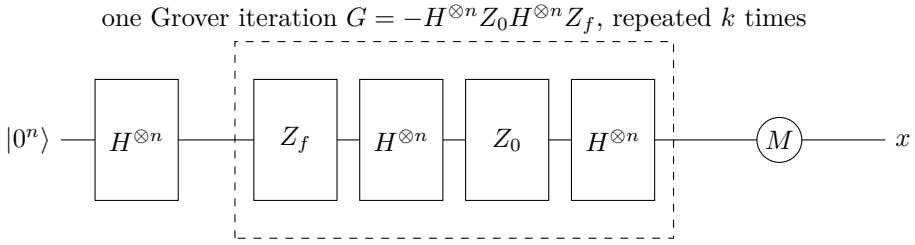


Figure 16: Grover's algorithm on n qubits (the extra qubit used by Z_f and Z_0 is not drawn). Each iteration makes one query.

30 Analysis: A Rotation in Two Dimensions

The exact analysis works for any number a of solutions, $1 \leqslant a \leqslant N-1$; let $b = N - a$. The key observation is that the whole computation takes place in a two-dimensional plane: one dimension for the solutions and one for the non-solutions. Let

$$|A\rangle = \frac{1}{\sqrt{a}}\sum_{\substack{x\in\{0,1\}^n \\ f(x)=1}}|x\rangle, \qquad |B\rangle = \frac{1}{\sqrt{b}}\sum_{\substack{x\in\{0,1\}^n \\ f(x)=0}}|x\rangle$$

be the uniform superpositions of the solutions and of the non-solutions. They are unit vectors (each has a, resp. b, amplitudes equal to $1/\sqrt{a}$, resp. $1/\sqrt{b}$) and they are orthogonal (no basis state appears in both). The initial state is

$$|h\rangle = \frac{1}{\sqrt{N}}\sum_{x\in\{0,1\}^n}|x\rangle = \frac{1}{\sqrt{N}}\Big(\sqrt{a}|A\rangle + \sqrt{b}|B\rangle\Big) = \sqrt{\frac{a}{N}}|A\rangle + \sqrt{\frac{b}{N}}|B\rangle. \tag{4}$$

Since $\frac{a}{N} + \frac{b}{N} = 1$, there is a unique angle $\theta \in (0, \pi/2)$ with

$$\sin\theta = \sqrt{\frac{a}{N}}, \qquad \cos\theta = \sqrt{\frac{b}{N}}, \qquad \text{so that} \qquad |h\rangle = \sin\theta|A\rangle + \cos\theta|B\rangle.$$

For a unique solution, $\sin\theta = 1/\sqrt{N}$, so $\theta \approx 1/\sqrt{N}$ is a small angle: $|h\rangle$ is almost $|B\rangle$ (Figure 17).

Lemma 31 ▶ One Grover Iteration

$$G|A\rangle = \left(1 - \frac{2a}{N}\right)|A\rangle - \frac{2\sqrt{ab}}{N}|B\rangle = \cos 2\theta|A\rangle - \sin 2\theta|B\rangle,$$
$$G|B\rangle = \frac{2\sqrt{ab}}{N}|A\rangle - \left(1 - \frac{2b}{N}\right)|B\rangle = \sin 2\theta|A\rangle + \cos 2\theta|B\rangle.$$

Proof. The phase flip Z_f negates every solution and fixes every non-solution, so $Z_f|A\rangle = -|A\rangle$ and $Z_f|B\rangle = |B\rangle$. By Equation (4) and orthonormality, $\langle h|A\rangle = \sqrt{a/N}$ and $\langle h|B\rangle = \sqrt{b/N}$. With $D = 2|h\rangle\langle h| - I$ (Lemma 30),

$$\begin{aligned} G|A\rangle = DZ_f|A\rangle = -D|A\rangle = -(2|h\rangle\langle h|A\rangle - |A\rangle) &= |A\rangle - 2\sqrt{\frac{a}{N}}|h\rangle \\ &= |A\rangle - 2\sqrt{\frac{a}{N}}\left(\sqrt{\frac{a}{N}}|A\rangle + \sqrt{\frac{b}{N}}|B\rangle\right) \\ &= |A\rangle - 2\left(\frac{a}{N}|A\rangle + \frac{\sqrt{ab}}{N}|B\rangle\right) = \left(1 - \frac{2a}{N}\right)|A\rangle - \frac{2\sqrt{ab}}{N}|B\rangle, \end{aligned}$$

and, by the same computation except that Z_f leaves $|B\rangle$ unchanged, so that the leading minus sign of $-D$ is not there,

$$\begin{aligned} G|B\rangle = DZ_f|B\rangle = D|B\rangle = 2|h\rangle\langle h|B\rangle - |B\rangle &= 2\sqrt{\frac{b}{N}}\left(\sqrt{\frac{a}{N}}|A\rangle + \sqrt{\frac{b}{N}}|B\rangle\right) - |B\rangle \\ &= \frac{2\sqrt{ab}}{N}|A\rangle + \frac{2b}{N}|B\rangle - |B\rangle = \frac{2\sqrt{ab}}{N}|A\rangle - \left(1 - \frac{2b}{N}\right)|B\rangle. \end{aligned}$$

For the trigonometric form use $\sin^2\theta = a/N$, $\cos^2\theta = b/N$ and the double angle formulas:

$$\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{2a}{N}, \qquad \cos 2\theta = 2\cos^2\theta - 1 = \frac{2b}{N} - 1 = -\left(1 - \frac{2b}{N}\right),$$
$$\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\sqrt{ab}}{N}.$$

□

(The $|B\rangle$ term of $G|A\rangle$ is negative; on the board it was written with a + sign in a few places, but the line before it, $|A\rangle - 2(\dots)$, and the matrix written in class both have the minus sign.) Since $a + b = N$, both $1 - 2a/N$ and $2b/N - 1$ equal $(b - a)/N$. So, writing a state $\beta|B\rangle + \alpha|A\rangle$ of the plane as the column vector $(\beta, \alpha)^T$ in the basis $(|B\rangle, |A\rangle)$, the lemma says that G acts as the 2×2 matrix

$$G = \begin{pmatrix} \frac{b-a}{N} & -\frac{2\sqrt{ab}}{N} \\ \frac{2\sqrt{ab}}{N} & \frac{b-a}{N} \end{pmatrix} = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, \tag{5}$$

whose columns are the coordinates of $G|B\rangle$ and $G|A\rangle$. The matrix on the right is the rotation matrix $R_\phi = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$ with $\phi = 2\theta$: one Grover iteration *rotates* the plane by the angle 2θ, from $|B\rangle$ towards $|A\rangle$. (Geometrically, Z_f is the reflection about the line of $|B\rangle$ and D is the reflection about the line of $|h\rangle$; two reflections about lines at angle θ compose to a rotation by 2θ.) In class the computation of $G|h\rangle$ was started and the rotation matrix was written down, but the computation was not finished; we complete it here.

Exercise 7. Compute $G|h\rangle = \sqrt{a/N}\,G|A\rangle + \sqrt{b/N}\,G|B\rangle$ and, more generally, $G^k|h\rangle$, and verify that the matrix of G in Equation (5) is the rotation $R_{2\theta}$.

Solution. *The matrix is a rotation.* By Lemma 31 and $(b-a)/N = 1 - 2a/N = 2b/N - 1$, the two columns of the matrix in Equation (5) are the coordinates of $G|B\rangle = \cos 2\theta|B\rangle + \sin 2\theta|A\rangle$ and $G|A\rangle = -\sin 2\theta|B\rangle + \cos 2\theta|A\rangle$, which are exactly the columns of $R_{2\theta}$. (It is $R_{2\theta}$, not R_θ: the entries are $\cos 2\theta = 1 - 2\sin^2\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$, not $\cos\theta$ and $\sin\theta$.) As a check, the columns are unit vectors and orthogonal: $((b-a)/N)^2 + \left(2\sqrt{ab}/N\right)^2 = (b^2 - 2ab + a^2 + 4ab)/N^2 = (a+b)^2/N^2 = 1$.

One iteration. By the trigonometric form of Lemma 31 and the addition formulas,

$$G|h\rangle = \sin\theta\,(\cos 2\theta|A\rangle - \sin 2\theta|B\rangle) + \cos\theta\,(\sin 2\theta|A\rangle + \cos 2\theta|B\rangle) = \sin 3\theta|A\rangle + \cos 3\theta|B\rangle.$$

(In the a, b form started in class, the coefficient of $|A\rangle$ is $\sqrt{a/N}\,(1 - 2a/N) + \sqrt{b/N}\cdot 2\sqrt{ab}/N = \sqrt{a/N}\cdot(3b-a)/N = \sin\theta(3\cos^2\theta - \sin^2\theta) = \sin 3\theta$ by the triple angle formula $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$.)

k iterations. We claim that for every $k \geqslant 0$,

$$G^k|h\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle.$$

This holds for $k = 0$ by Equation (4). If it holds for k, then by Lemma 31 and the addition formulas, with $\phi = (2k+1)\theta$,

$$\begin{aligned} G^{k+1}|h\rangle &= \sin\phi(\cos 2\theta|A\rangle - \sin 2\theta|B\rangle) + \cos\phi(\sin 2\theta|A\rangle + \cos 2\theta|B\rangle) \\ &= (\sin\phi\cos 2\theta + \cos\phi\sin 2\theta)|A\rangle + (\cos\phi\cos 2\theta - \sin\phi\sin 2\theta)|B\rangle \\ &= \sin(\phi + 2\theta)|A\rangle + \cos(\phi + 2\theta)|B\rangle = \sin((2k+3)\theta)|A\rangle + \cos((2k+3)\theta)|B\rangle, \end{aligned}$$

which is the claim for $k+1$. So after k iterations the state makes the angle $(2k+1)\theta$ with $|B\rangle$: it starts at θ and every iteration adds 2θ (Figure 17). □

30.1 How Many Iterations

Measuring the state $G^k|h\rangle = \sin((2k+1)\theta)|A\rangle + \cos((2k+1)\theta)|B\rangle$ in the computational basis gives a solution with probability $\sin^2((2k+1)\theta)$: the amplitude of each of the a solutions is $\sin((2k+1)\theta)/\sqrt{a}$, and these a squared amplitudes add up to $\sin^2((2k+1)\theta)$. We want $(2k+1)\theta$ to be as close to $\pi/2$ as possible.

Theorem 32 ▶ Grover's Algorithm

Let f have a solutions, $1 \leqslant a \leqslant N/2$, let $\theta \in (0, \pi/2)$ with $\sin\theta = \sqrt{a/N}$, and let $k = \lfloor \pi/4\theta \rfloor$. Then Grover's algorithm with k iterations makes $k \leqslant \frac{\pi}{4}\sqrt{N/a}$ queries and outputs a solution with probability at least $1 - a/N \geqslant 1/2$. For a unique solution ($a = 1$) it makes about $\frac{\pi}{4}\sqrt{N}$ queries and succeeds with probability at least $1 - 1/N$.

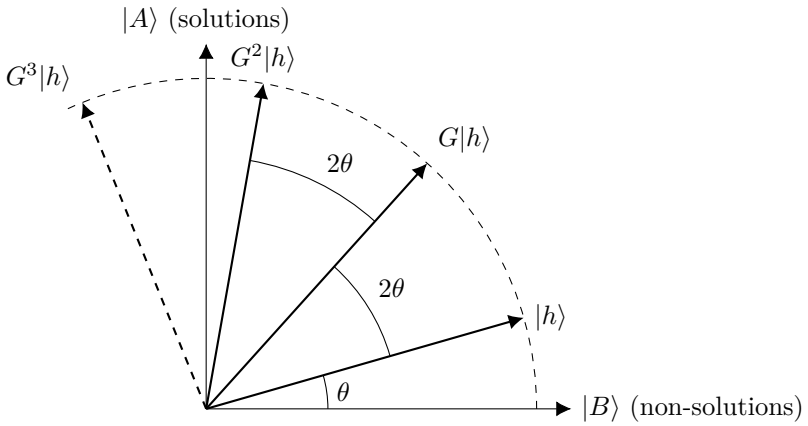


Figure 17: Grover's algorithm in the plane spanned by $|A\rangle$ and $|B\rangle$. The initial state $|h\rangle$ makes the small angle θ with $|B\rangle$, and every iteration rotates the state by 2θ towards $|A\rangle$: after k iterations the angle is $(2k+1)\theta$. Here $\theta = 16°$, so $k = \lfloor \pi/4\theta \rfloor = 2$ iterations are best (angle $80°$, success probability $\sin^2 80° \approx 0.97$), and a third one (dashed, angle $112°$) overshoots past $|A\rangle$ and lowers the probability to $\sin^2 112° \approx 0.86$.

Proof. By the definition of k, $\pi/4\theta - 1 < k \leqslant \pi/4\theta$, so multiplying by 2θ and adding θ,

$$\frac{\pi}{2} - \theta < (2k+1)\theta \leqslant \frac{\pi}{2} + \theta.$$

The sine is increasing on $[0, \pi/2]$ and decreasing on $[\pi/2, \pi]$, so on the interval $[\pi/2-\theta, \pi/2+\theta] \subseteq [0, \pi]$ it is at least its value at the endpoints, $\sin(\pi/2 \pm \theta) = \cos\theta$. Hence the success probability is

$$\sin^2((2k+1)\theta) \geqslant \cos^2\theta = 1 - \sin^2\theta = 1 - \frac{a}{N} \geqslant \frac{1}{2}.$$

For the number of queries, $\theta \geqslant \sin\theta = \sqrt{a/N}$ (as $\sin\theta \leqslant \theta$ for $\theta \geqslant 0$), so $k \leqslant \pi/4\theta \leqslant \frac{\pi}{4}\sqrt{N/a}$. For $a = 1$, $\sin\theta = 1/\sqrt{N}$, so $\theta \approx 1/\sqrt{N}$ and $k \approx \frac{\pi}{4}\sqrt{N}$. □

This is the precise form of "repeat $\sqrt{N}$ times": the number of iterations is $\Theta\left(\sqrt{N}\right)$ for a unique solution and $\Theta\left(\sqrt{N/a}\right)$ for a solutions, and the success probability is a constant close to 1. The analysis also explains the two-qubit example: for $N = 4$ and $a = 1$, $\sin\theta = 1/2$, so $\theta = \pi/6$ and $k = \lfloor 3/2 \rfloor = 1$ iteration gives the angle $3\theta = \pi/2$, i.e., the state $|A\rangle = |x^*\rangle$ exactly and success probability $\sin^2(\pi/2) = 1$, as computed in Figure 14.

Do not overshoot. Unlike a classical search, more iterations do not help: the state keeps rotating, and after k iterations with $(2k+1)\theta > \pi/2$ the success probability $\sin^2((2k+1)\theta)$ decreases again (the dashed vector in Figure 17). After about $\pi/2\theta$ iterations the state is back near $|B\rangle$. So the number of iterations has to be chosen from a and N; this is why the algorithm needs to know (or to estimate) the number of solutions.